Physical Limits in Photonics: From Radiative Heat Transfer to Information Capacity and Design

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Prof. A. Rodriguez



T. Maldonado



F.J. Chen



Dr. B. Strekha



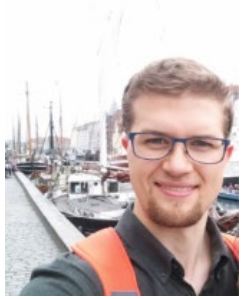
Dr. J. Mohajan



Collaborators

@ Polytech Montreal

Prof. S. Molesky



P. Virally



@ Virginia Tech

Prof. Z. Lin



@ MIT

Dr. P. Chao

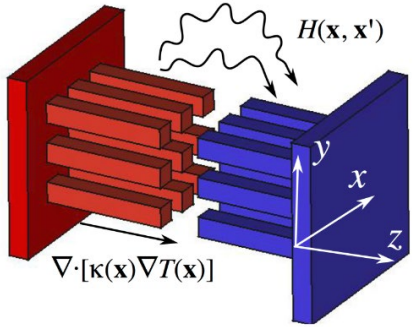


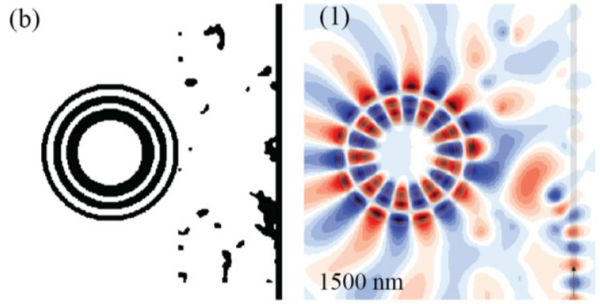
Prof. S. Johnson

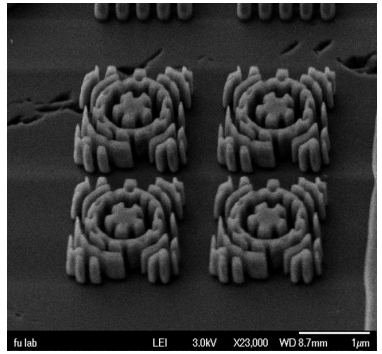


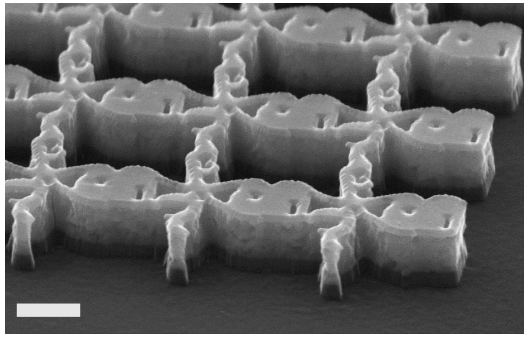
In our group

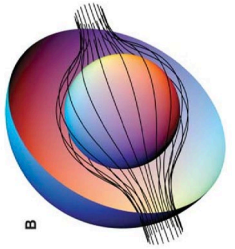
- Large-scale optimization (design)
- Heat transfer (cooling, near field physics)
- Nonlinear optics
- Numerical methods + leveraging ML infrastructure for performance
- Quantum fluctuations (Casimir forces, MEMS, single-photon photodetectors)
- Communication (information theory)
- Wave transformations (signal processing, imaging, cloaking, steering, focusing)

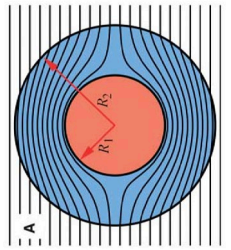




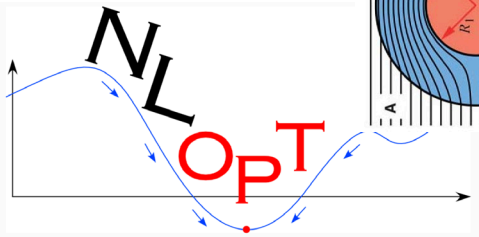










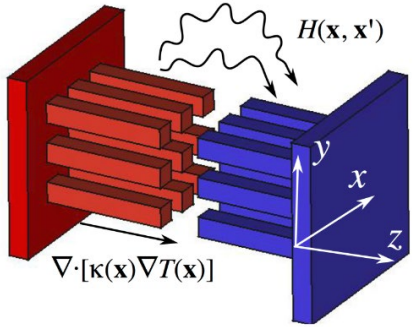


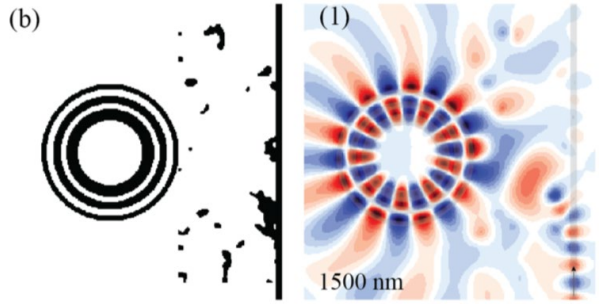


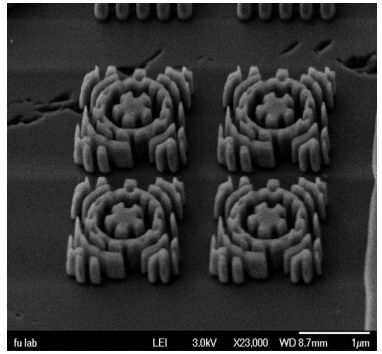
<u>+ physical limits on all these quantities</u>

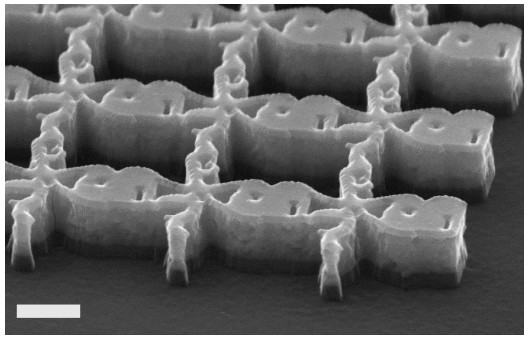
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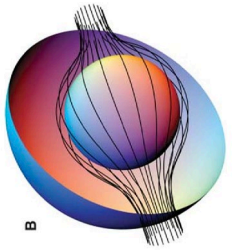
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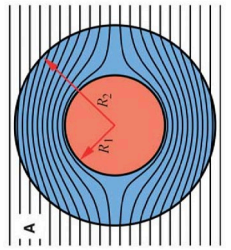




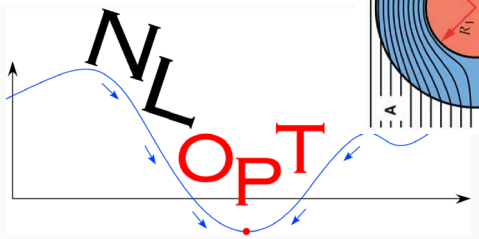








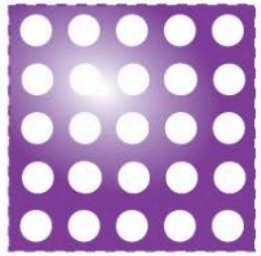




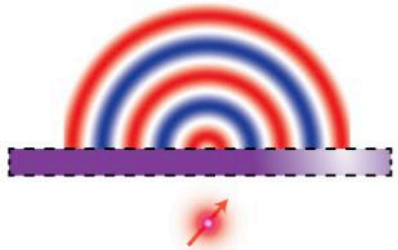


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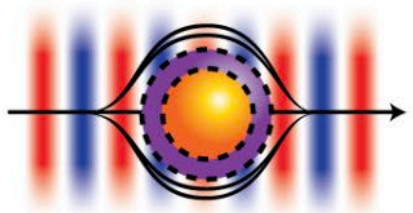
Many Open Structural Design Problems in Electromagnetism



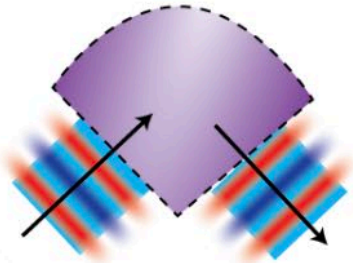
confinement



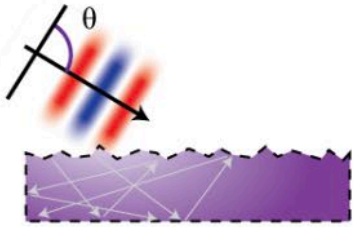
radiative enhancement



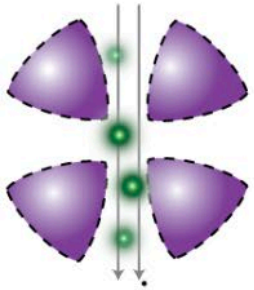
obscurance



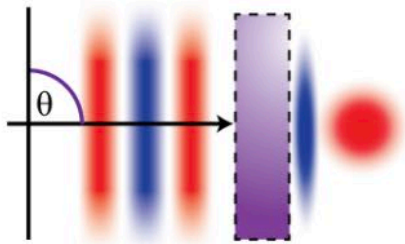
coupling



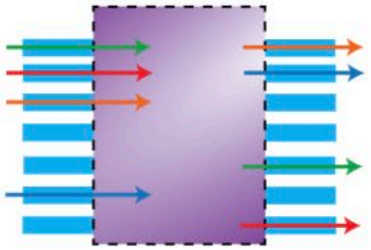
light trapping



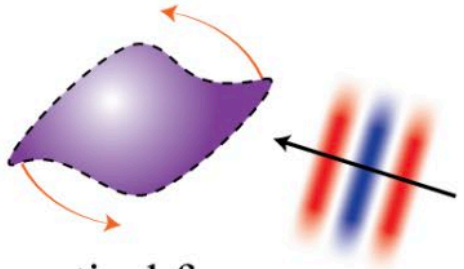
sensing



lensing



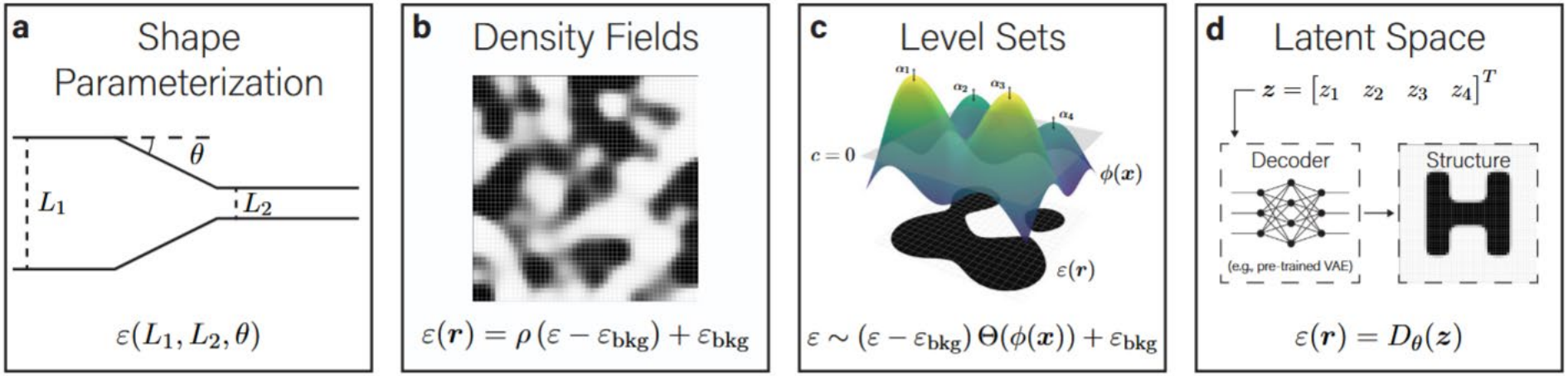
communication



optical forces

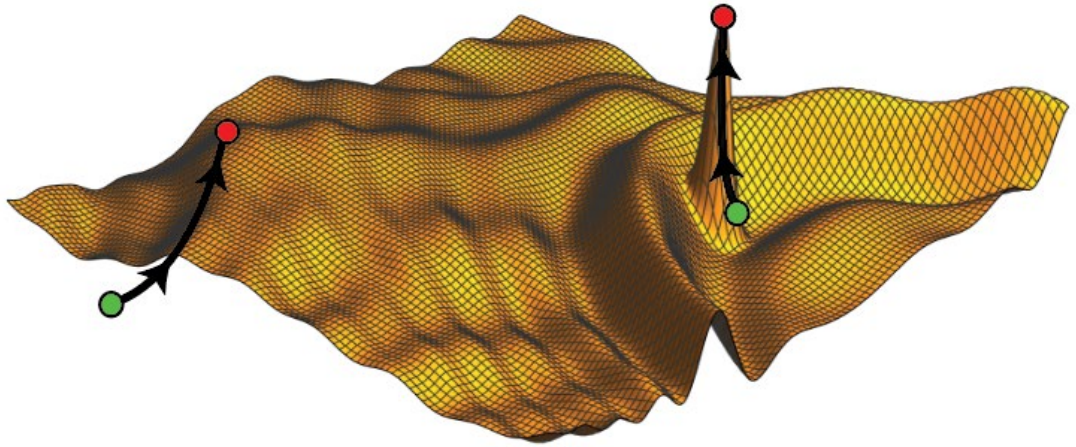
It is not obvious how to do these well!
Or all the ways *to* do these

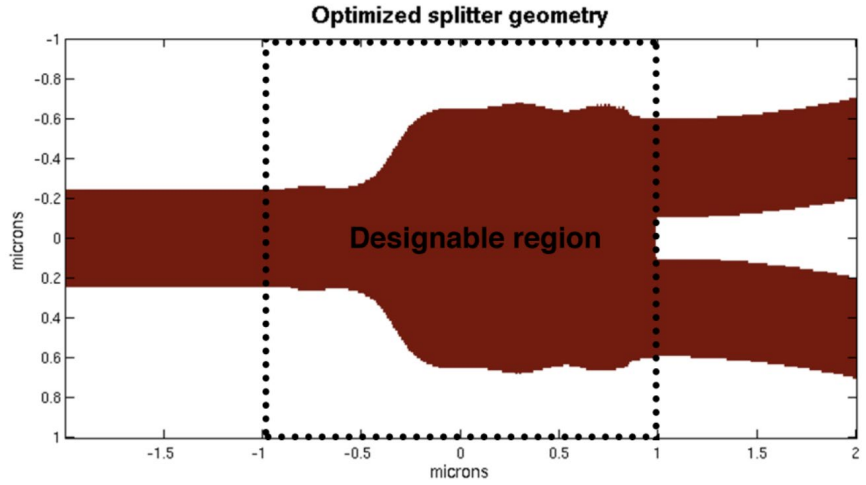
What is inverse design?



Very high-dimensional!

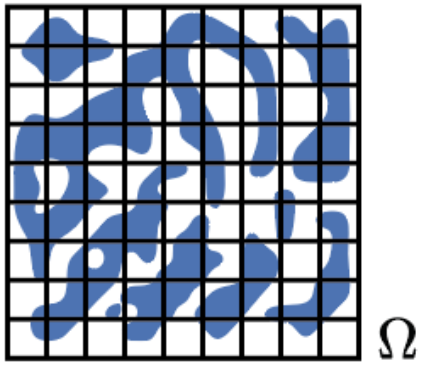
Yablonovitch 2013





How do you do inverse design?

inverse design



$$\max f(\boldsymbol{E}_t) \quad \text{s.t.} \quad \chi(\boldsymbol{r}) \in \Omega$$

$$\nabla \times \nabla \times \boldsymbol{E}_t - (\mathbf{1} + \chi(\boldsymbol{r}))\omega^2 \boldsymbol{E}_t = i\omega \boldsymbol{J}_i$$

Maxwell's equations

$$\max_{\psi,\rho} \quad \mathcal{F}[\psi(\mathbf{r}), \psi^*(\mathbf{r}), \rho(\mathbf{r})]$$

$$\text{s.t.} \quad \mathcal{M}[\psi(\mathbf{r}), \rho(\mathbf{r})] = 0,$$

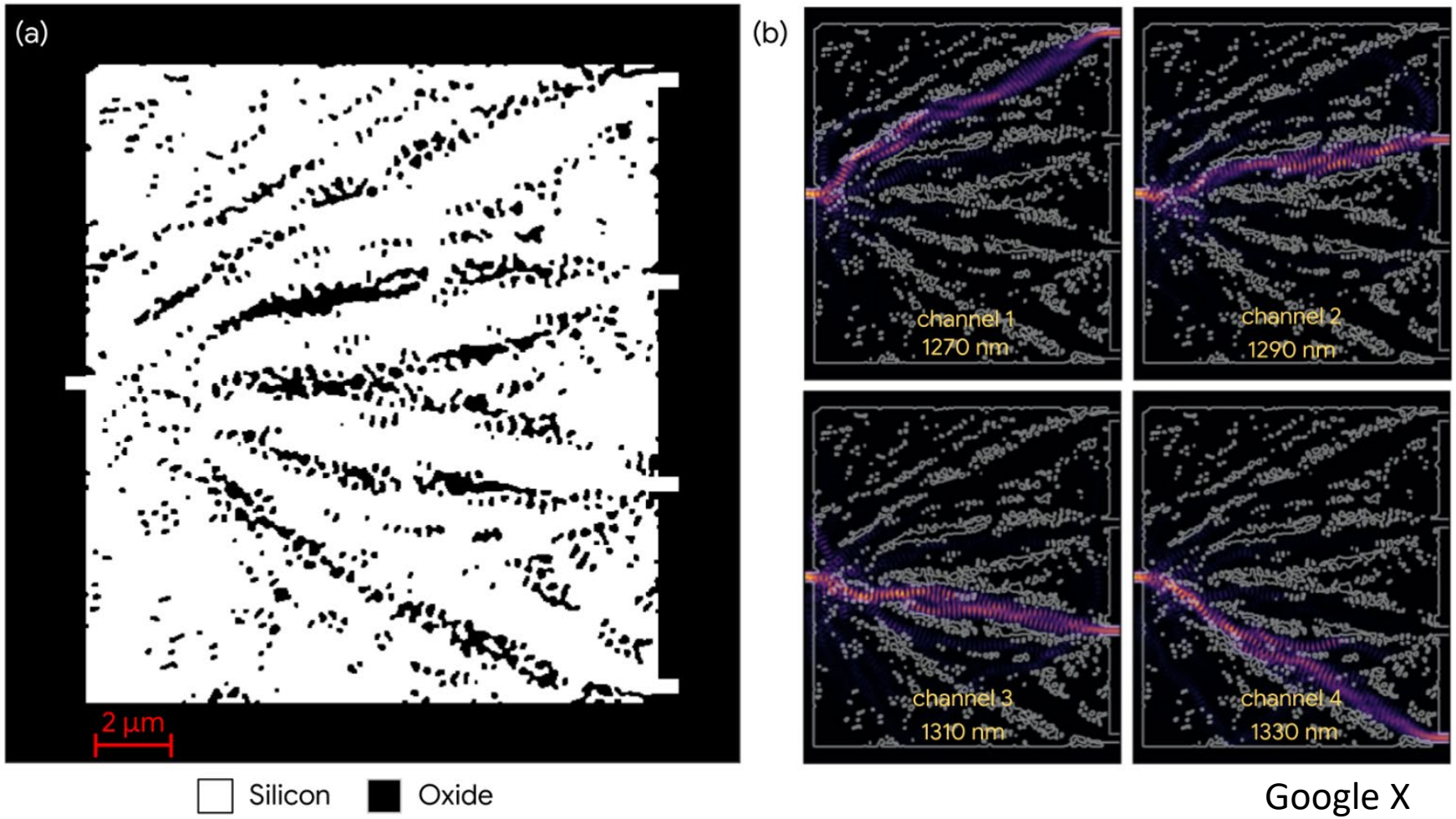
$$\frac{d\mathcal{F}}{d\rho(\mathbf{r})} = \frac{\partial \mathcal{F}}{\partial \rho(\mathbf{r})} + 2\,\mathrm{Re}\left(\frac{\partial \mathcal{F}}{\partial \psi(\mathbf{r})}\frac{\partial \psi(\mathbf{r})}{\partial \rho(\mathbf{r})}\right),$$

The way to get these gradients is often called the adjoint method

it is the same process as backwards differentiation in ML

(be wary of claims of auto-differentiation being faster or more efficient than the "traditional adjoint method": the adjoint method **is** the differentiation rule for linear solves)

Inverse design is very successful at device engineering!



Google X

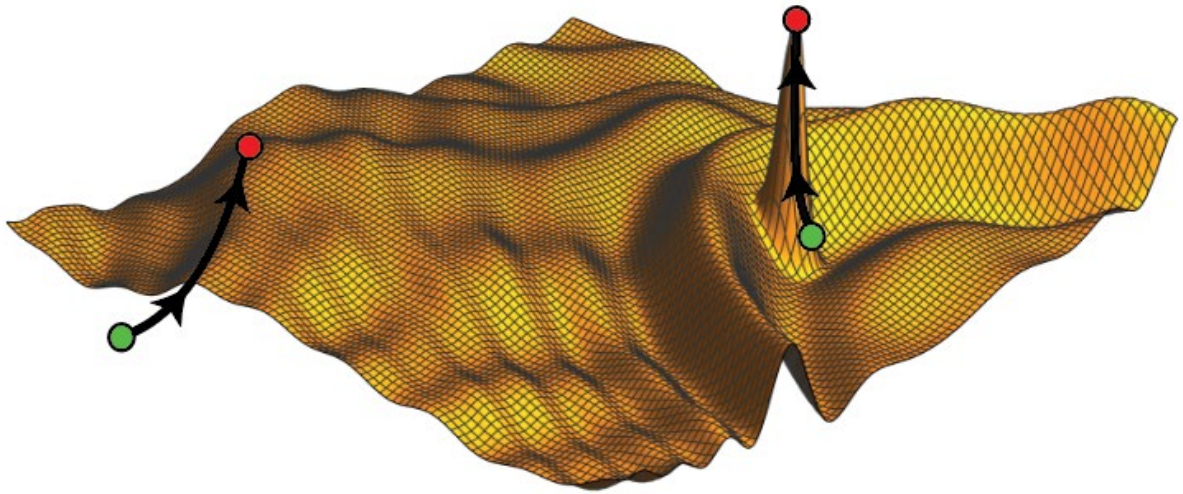
Structures are often very high performing and very compact

Some groups who are really good at inverse design: Ole Sigmund, Eli Yablonovitch, Jelena Vuckovic, Rasmus Christiansen, Steven Johnson, Shanhui Fan, and more!

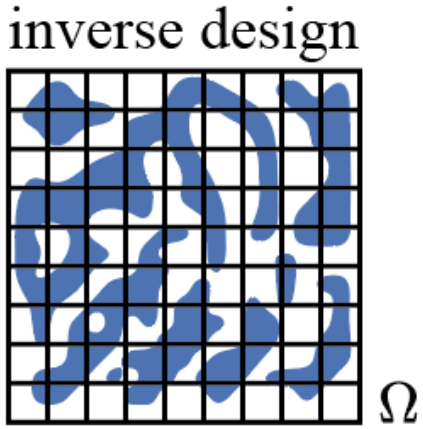
The design challenge

Lots of **degrees of freedom** in geometry, but **what structure is best** for given device & materials?

And what **performance & phenomena are possible?**



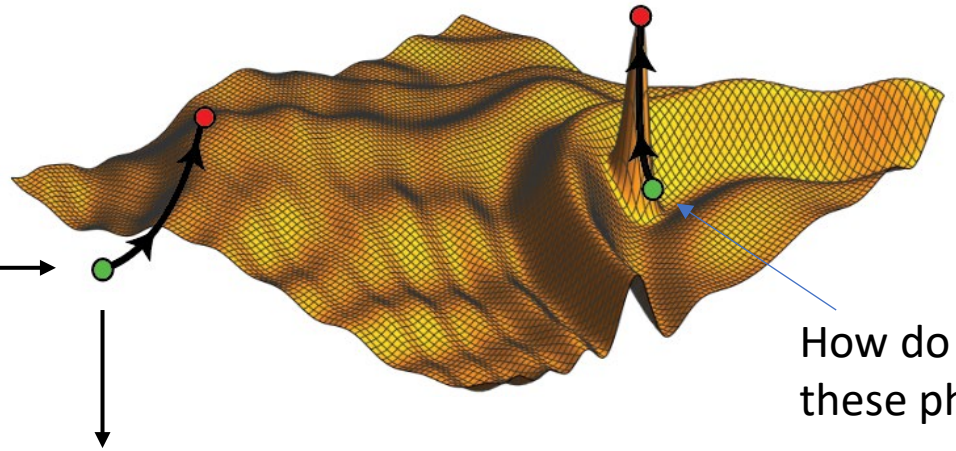
Inverse design is NP-hard



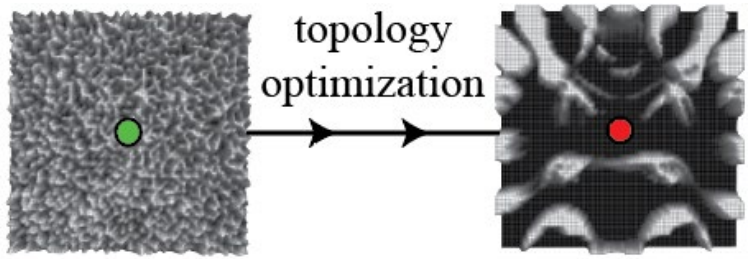
$$\max f(\boldsymbol{E}_t) \quad \text{s.t.} \quad \chi(\boldsymbol{r}) \in \Omega$$

$$\nabla \times \nabla \times \boldsymbol{E}_t - (1 + \chi(\boldsymbol{r}))\omega^2 \boldsymbol{E}_t = i\omega \boldsymbol{J}_i$$

Maxwell's equations



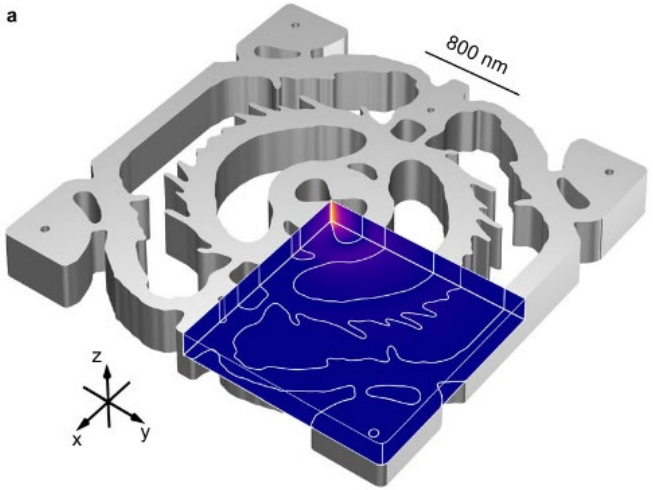
How do we access these physics?



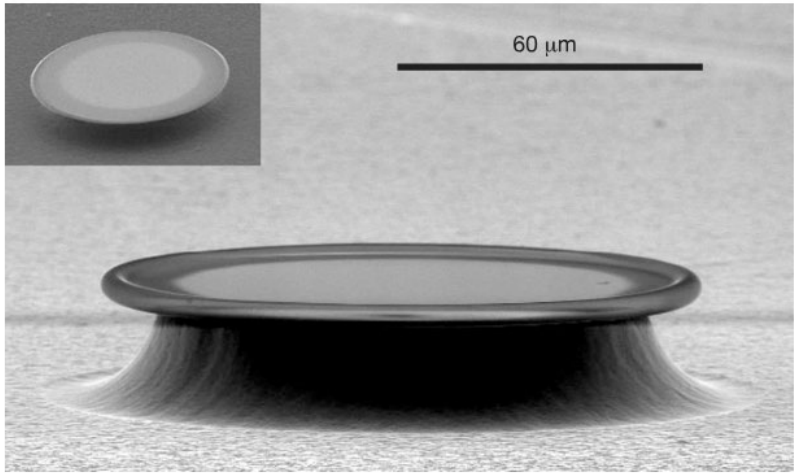
Every pixel's refractive index is a continuous degree of freedom $\varepsilon(r) \in [\varepsilon_{min}, \varepsilon_{max}]$

Many successful applications! But likely local maxima

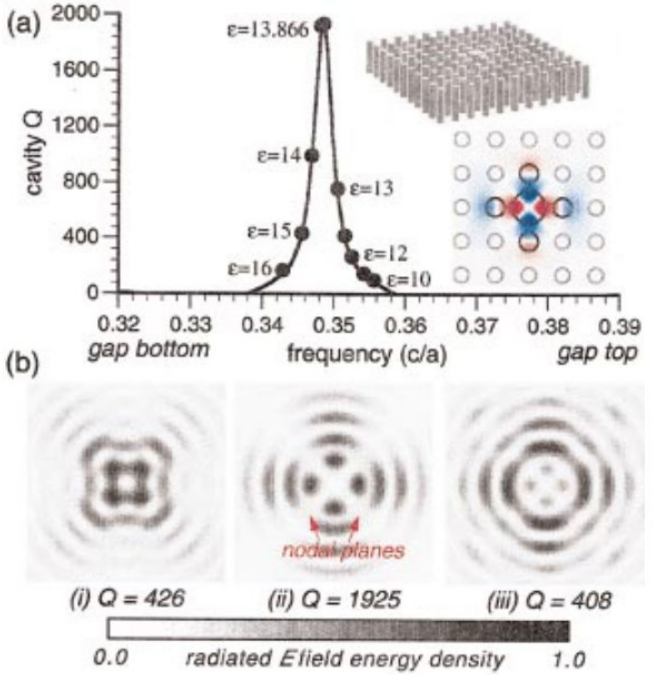
Why can inverse design / topology optimization fail?



Localized Fields
(gradients near zero)



Resonance Engineering
(sensitive to perturbations)

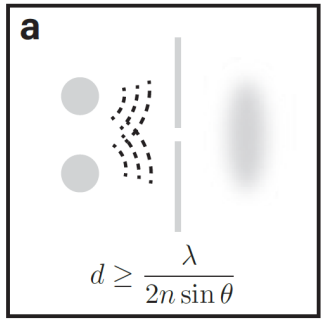


Interference Engineering
(very sensitive to perturbations)

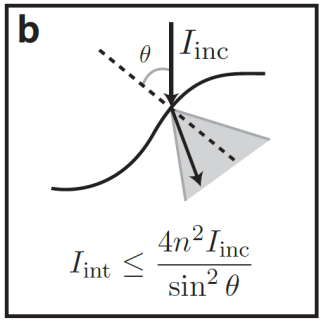
Other reasons: just simply stuck, curse of dimensionality, parametrization is bad, bad initialization, etc.

Motivates understanding the best possible performance

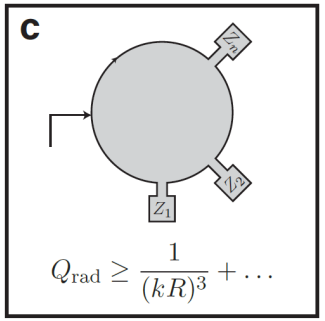
Limits in Photonics



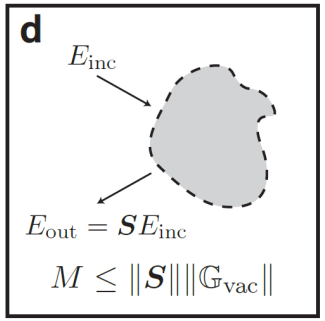
Abbe Diffraction



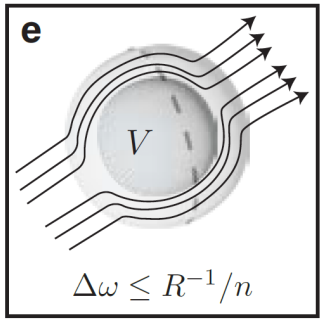
Light Trapping



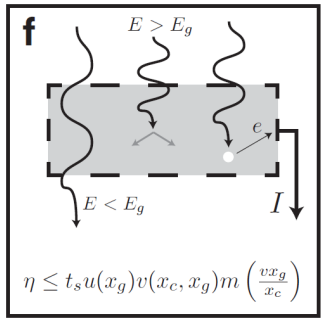
Antenna Radiation



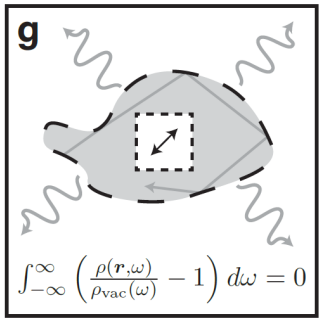
Communication



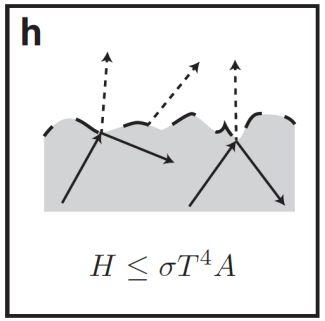
Cloaking



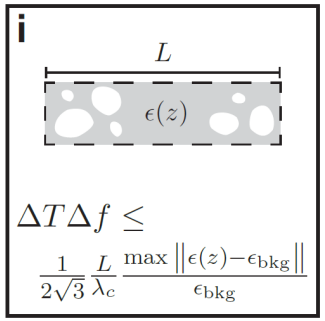
Photovoltaic Efficiency



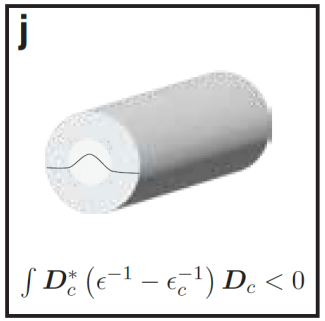
Light Extraction



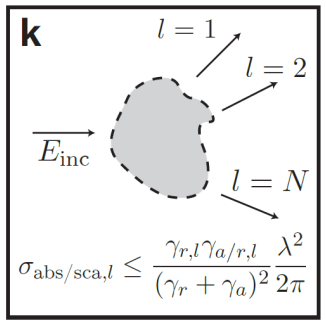
Blackbody Radiation

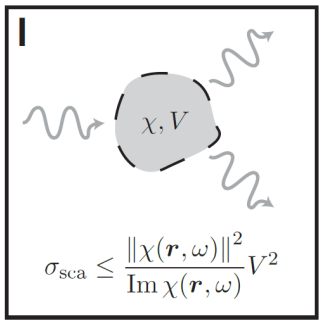


Delay-Bandwidth

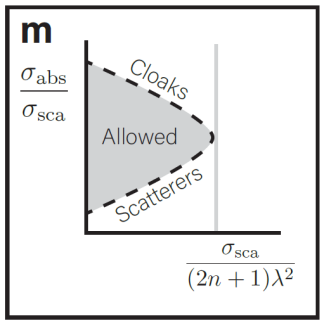


Waveguide Confinement

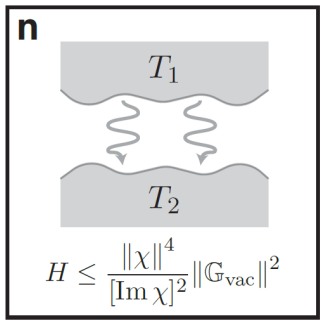




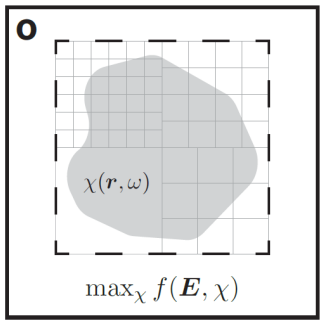
Absorption & Scattering Cross-Sections



Extinction



Heat Transfer



General Limits?

What do limits tell us?

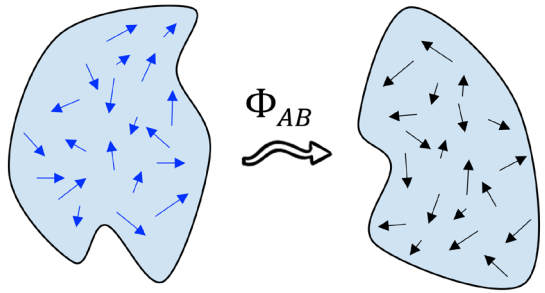
Typically, we know materials, but don't know the structure, so geometry-agnostic & material-dependent limits are especially useful for guiding design.

Performance bounds can:

1. serve as targets for design (certificates of optimality)
2. Inform scaling behavior (with size, material, etc.)
3. Seed inverse design for optimization
4. Inform design via new physics

Dependency Graph / Outline

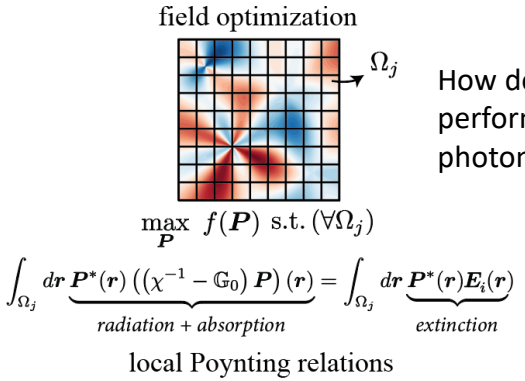
1. Back to heat transfer



Sum of emergent channels

Limits on heat transfer

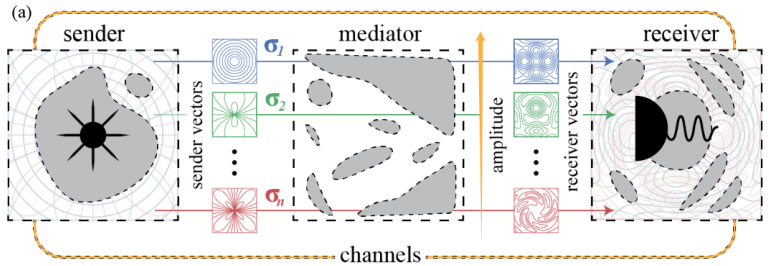
Limits in Photonics



How do we compute performance bounds on photonic devices?

Same!

3. Information Theory: Communication + Sensing



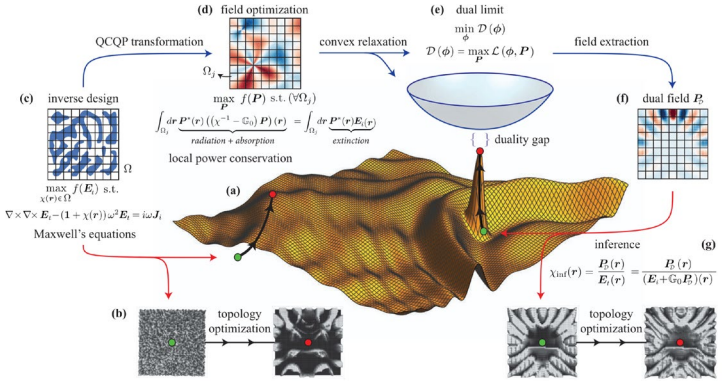
Formulate photonic Shannon capacity

Clear notion of **information channels**

Limits on photonic communication and sensing

Revisit heat transfer limits?

4. Design from limits



Formulate how to use information in limits to initialize optimization

Show off some insane, impossible cavities

$\chi = 2\,, Q = 10^{12}$

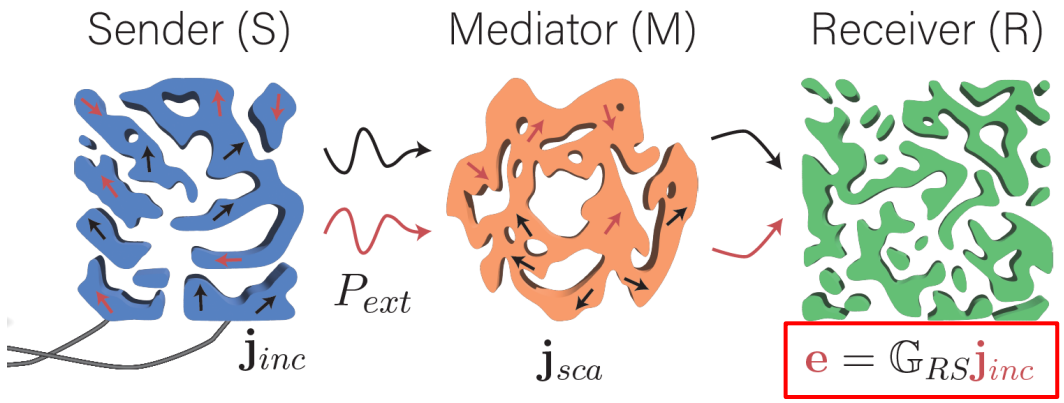
Part 1:

Heat Transfer Between Structured Bodies

&

General Fundamental Limits in Photonics

Photonic “Channels”



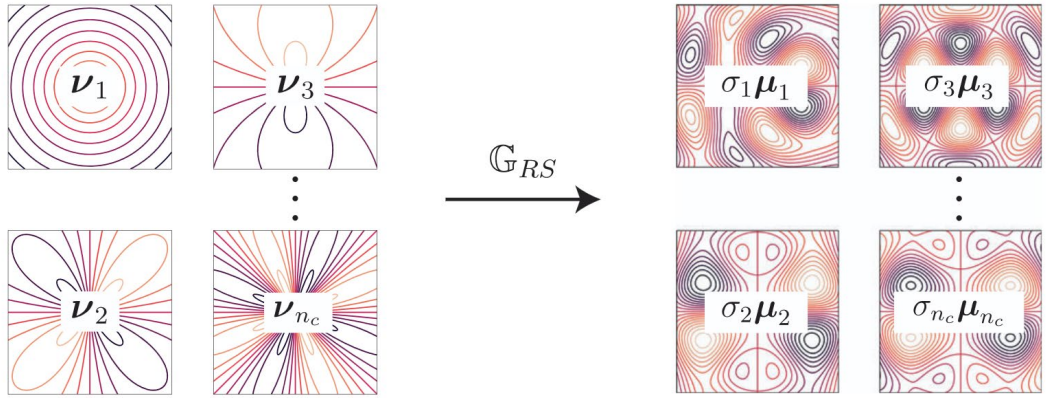
$$\mathbb{G}_{RS} = \sum_j \sigma_j u_j v_j^\dagger$$

Singular vectors are the “modes”
Each mode pair is a “channel”

See work by:
- David Miller
- Owen Miller
- Mats Gustafsson
- Alejandro Rodriguez

$$\mathbb{G}_{RS} = \sum_j \sigma_j \boldsymbol{\mu}_j \boldsymbol{\nu}_j^\dagger$$

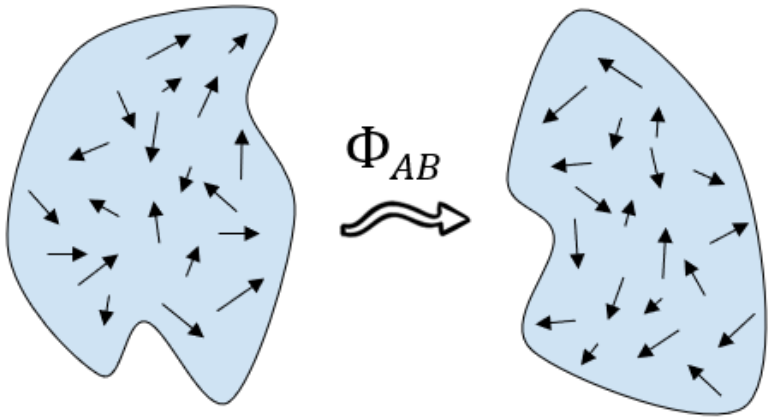




Natural mathematical decomposition into sender and receiver modes
(will return as communication channels; for now just photonic channels)

Heat Transfer between Structured Bodies

$$H = \int_0^\infty \left(\Pi(\omega, T_H) - \Pi(\omega, T_C)\right) \cdot \Phi(\omega)\, \mathrm{d}\omega$$



Field from each dipole

$$|\mathbf{E}|^2 \sim \frac{k^6}{8\pi^2}\left[\frac{3}{(kr)^6} + \frac{1}{(kr)^4} + \frac{1}{(kr)^2}\right]$$

Near-field contributions

Blackbody Limits:

- Ray optics
- Perfect Emission & Absorption
- No material dependence

Fluctuation density — Material dissipation

$$\langle \mathbf{J}(\mathbf{x}, \omega)\mathbf{J}(\mathbf{x}', \omega)\rangle = \frac{\omega}{\pi}\,\mathrm{Im}[\varepsilon(\mathbf{x}, \omega)]\,\Pi(T(\mathbf{x}), \omega)\delta(\mathbf{x}' - \mathbf{x})$$

$$\Phi = \left\langle \int_{V_A}\int_{V_B} (\mathrm{Im}\chi_B)|\mathbf{E}|^2 \right\rangle$$

$$= \int_{V_A}\int_{V_B} (\mathrm{Im}\chi_B)|\mathbb{G}|^2 \langle \mathbf{JJ}\rangle$$

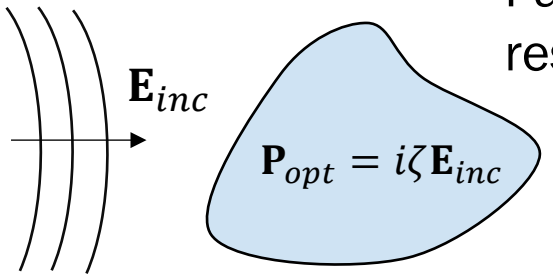
$$= \frac{2}{\pi}\int_{V_A}\int_{V_B} (\mathrm{Im}\chi_A)(\mathrm{Im}\chi_B)|\mathbb{G}|^2$$

$$= \frac{2}{\pi\zeta_A\zeta_B}\left\|\mathbb{G}_{t,BA}\right\|_F^2 = \sum_j \sigma_j^2$$

Prior limits: bounding $\|\mathbb{G}_{t,BA}\|_F^2$

Real slabs $\sim \frac{\log(\zeta_A\zeta_B)}{d^2}$

- Attempt 1: Passivity and reciprocity



Passivity: maximum polarization response bounded by loss

Miller, O.D., Johnson, S.G., Rodriguez, A.W. *Shape-Independent Limits to Near-Field Radiative Heat Transfer.* 2015.

$$\|\mathbb{T}\|^2 \le \zeta\|\mathbf{E}_{inc}\|^2$$

$$\Phi = \frac{2}{\pi\zeta_A\zeta_B}\underbrace{\|\mathbb{Y}_B\mathbb{G}_{BA}^{vac}\mathbb{T}_A\|_F^2}_{} \le \frac{2}{\pi}\zeta_A\zeta_B\sum_j\left(\sigma_j^{vac}\right)^2 \sim \frac{\zeta_A\zeta_B}{d^2}$$

$$\|\mathbb{Y}_B\mathbb{G}_{BA}^{vac}\mathbb{T}_A\|_F^2 \le \|\mathbb{Y}_B\|^2\|\mathbb{G}_{BA}^{vac}\|_F^2\|\mathbb{T}_A\|^2 \quad \text{(Cauchy–Schwarz)}$$

$$\|\mathbb{Y}_B\|^2 \le \|\mathbb{T}_B\|^2 \quad \text{(Perturbative "Born" response)}$$

$$\|\mathbb{T}_{A,B}\|^2 \le \zeta_{A,B} \quad \text{(passivity)}$$

- Attempt 2: Algebraic relaxations

$$\mathbb{Y}_B = \sum_j y_j|\mathbf{b}_j^*\rangle\langle\mathbf{b}_j|$$

$$\mathbb{T}_A = \sum_j \tau_j|\mathbf{a}_j^*\rangle\langle\mathbf{a}_j|$$

$$\mathbb{G}_{BA}^{vac} = \sum_j \sigma_j|\mathbf{b}_j^*\rangle\langle\mathbf{a}_j|$$

- Venkataram, P. S., et al. *Channel-based algebraic limits to conductive heat transfer*
- Venkataram, P.S., et al. *Fundamental Limits to Radiative Heat Transfer: The Limited Role of Nanostructuring in the Near-Field*
- Molesky, S. et al. *Fundamental limits to radiative heat transfer: Theory*
- Neumann, J.V. *Some matrix inequalities and metrization of matric space. 1937.*

$$\Phi = \frac{2}{\pi\zeta_A\zeta_B}\|\mathbb{Y}_B\mathbb{G}_{BA}^{vac}\mathbb{T}_A\|_F^2$$

$$\le \frac{2}{\pi\zeta_A\zeta_B}\sum_j\underbrace{\left(y_j\sigma_j^{vac}\tau_j\right)^2}_{}$$

Saturates at Landauer limit

$$\underbrace{y_j \le \frac{\zeta_B}{1+\zeta_A\zeta_B\left(\sigma_j^{vac}\right)^2} \qquad \tau_j \le \zeta_A}_{\text{(passivity)}}$$

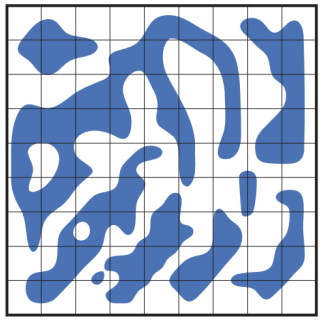
- Yields sum over *dressed* channel amplitudes
  - No longer each body in isolation
- *Relaxation*: each channel maximized independently
- Bound on $y_j\sigma_j\tau_j$ <u>not</u> a bound on singular values of $\mathbb{G}_{t,BA}$

$$\sim \frac{\log^2(\zeta_A\zeta_B)}{d^2}$$

Can we do better? Recent field relaxation framework

a

Structural Optimization

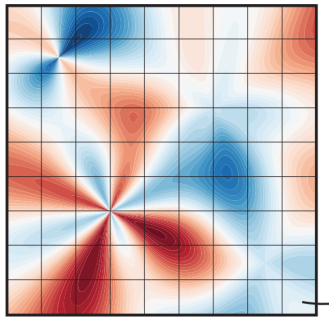


$$\max f(\mathbf{E})$$
$$\text{s.t.} \quad [\nabla \times \nabla \times -\epsilon(\mathbf{r})\omega^2]\mathbf{E}(\mathbf{r}) = i\omega\mathbf{J}$$

Maxwell's Equations

$$\epsilon(\mathbf{r};\omega) = \begin{cases} 1 \text{ or } 1+\chi(\omega) & \mathbf{r} \in V_{design} \\ 1 & \mathbf{r} \notin V_{design} \end{cases}$$

Field Optimization



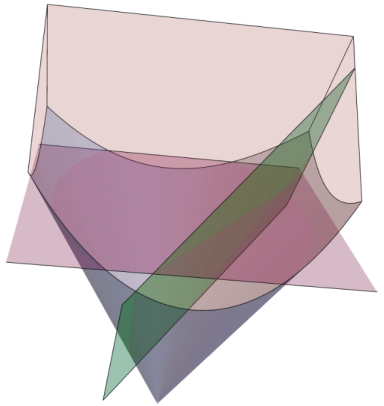
$$\max f(\epsilon(\mathbf{r}), \mathbf{E}(\mathbf{r}))$$
$$\int_{\Omega_j} \mathbf{P}^* \mathbf{E}_i = \int_{\Omega_j} \mathbf{P}^* \left(\chi^{-\dagger}(\mathbf{r}) - \mathbb{G}_0^{\dagger}\right) \mathbf{P}$$

$P_{in} = P_{abs} + P_{out}$

Local Poynting's Theorem (PT) → Convexify via duality

b

Semidefinite Programming



$$\begin{aligned} \max_{\mathbb{X}} \quad & \text{Tr}\,(\mathbb{D}_0\mathbb{X}) \\ \text{s.t.} \quad & \text{Tr}\,(\mathbb{D}_j\mathbb{X}) = b_j \\ & \mathbb{X} \succcurlyeq 0 \\ & \text{rank}(\mathbb{X}) = 1 \end{aligned}$$

→ Convexify via rank-relaxation

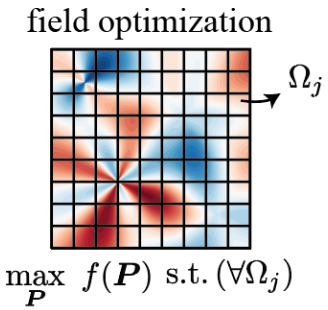
Same bound

Big idea: bounds on arbitrary structures given **size** & **material**

A few nice reviews:
Chao, P. et al. Physical limits in electromagnetism (2022)
Miller, O. D., Monticone, F. Fundamental limits in photonics and electromagnetics: a tutorial (2026)
Amaolo, A., et al. From inverse design to fundamental limits: new opportunities for photonic design (In preparation)

See also related works by
Owen Miller (Yale)
Steven G Johnson (MIT)
Mats Gustafsson (Lund U.)
Miloslav Capek (CTU Prague)
Rasmus Christiansen (DTU)
Philipp del Hougne (Rennes)

General limits as a generalization of absorbed power limits (related problem)



$\underset{\mathbf{P(r)}}{\text{maximize}} \; f(\mathbf{P(r)})$

Field relaxation – absorbed power s.t. to **passivity**

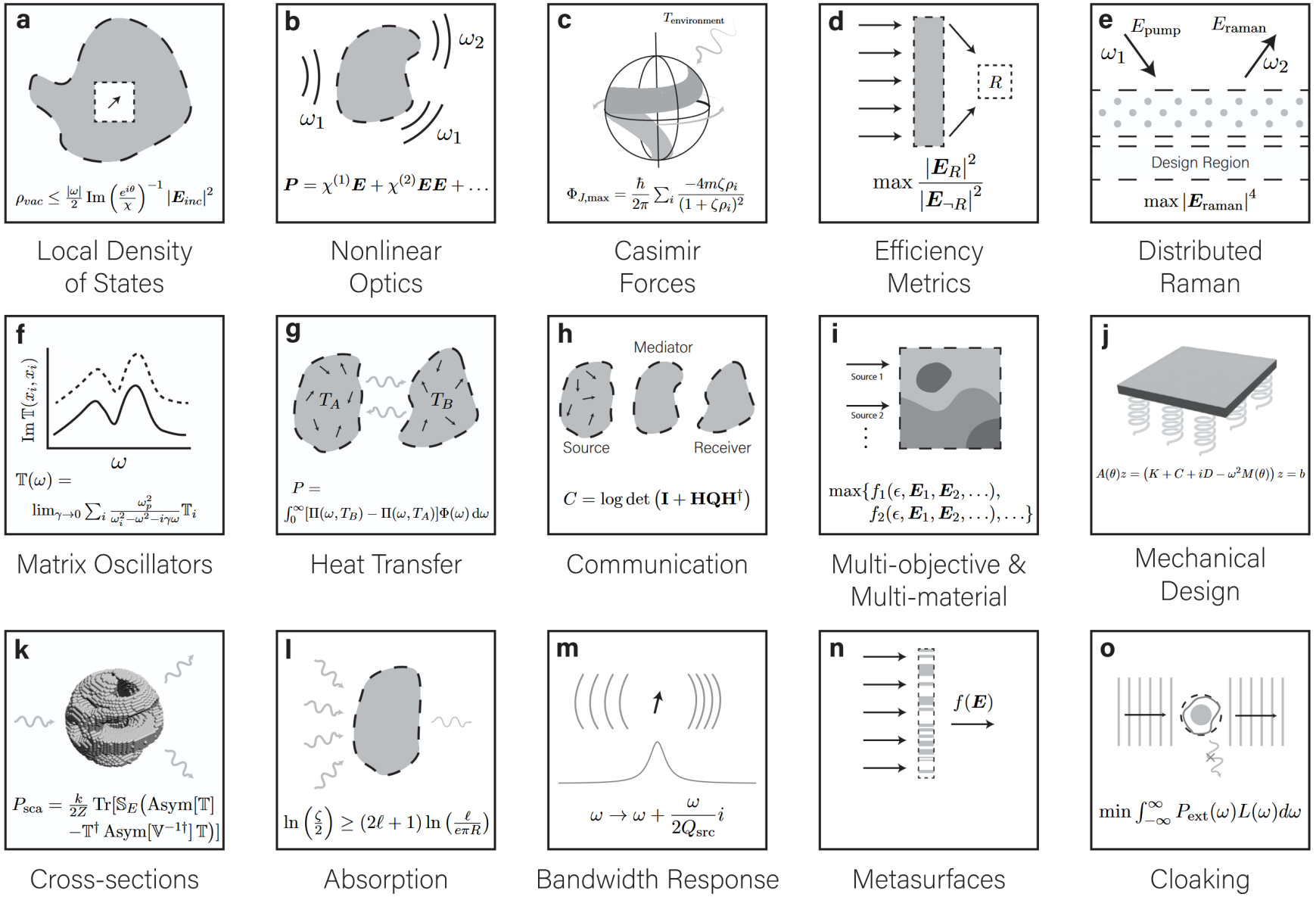
Get the same result

$$\Phi_{\text{opt}} = \sum_i \frac{1}{2\pi}\Theta(2\zeta\rho_i - 1) + \sum_i \frac{2}{\pi}\zeta\rho_i(1-\zeta\rho_i)\,\Theta(1-2\zeta\rho_i)$$

Emission into vacuum (also s.t. passivity)

New constraints (energy conservation) tighten these limits by including more PDE information

Molesky, S., Jin, W., Venkataram, P.S., Rodriguez, A.W., T Operator Bounds on Angle-Integrated Absorption and Thermal Radiation for Arbitrary Objects



[a]Chao, P. et al. *Maximum electromagnetic local density of states via material structuring*
[b]Amaolo, A. et al. *Physical Limits on Raman Scattering: The Critical Role of Pump and Signal Co-design*
[h]Amaolo, A. and Chao, P., et al. *Maximum Shannon Capacity of Photonic Structures*
[i]Amaolo, A. et al. *Can photonic heterostructures provably outperform single-material geometries?*
[k]Molesky, S. et al. *Global T operator bounds on electromagnetic scattering: Upper bounds on far-field cross sections*
[n]Molesky, S. et al. *T-operator limits on optical communication: Metaoptics, computation, and input-output transformations*

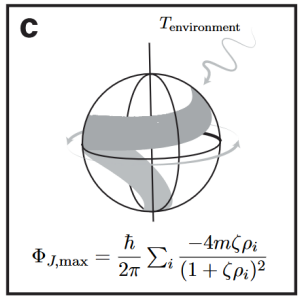
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Increasing antenna bandwidth and quantum emission

Local Density of States

Giant enhancement of Raman spectroscopy

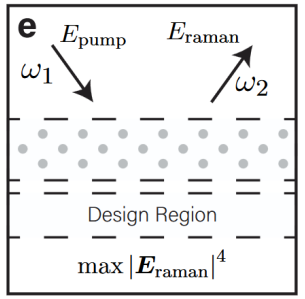
Nonlinear Optics



Casimir Forces

How much can light be focused?

Efficiency Metrics



Distributed Raman

What structures maximize near-field radiative heat transfer?

Matrix Oscillators

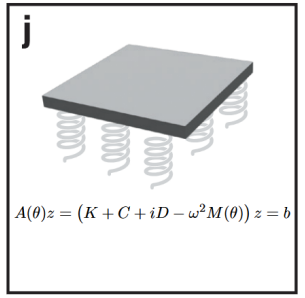
Heat Transfer

Enhancement of Shannon Capacity

Communication

When to design with multiple materials?

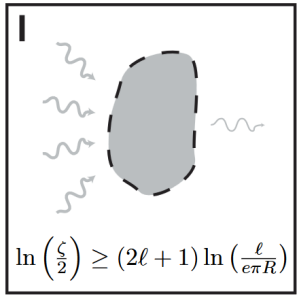
Multi-objective & Multi-material



Mechanical Design

How much can light be scattered?

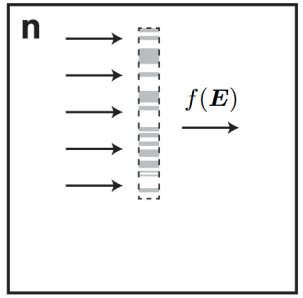
Cross-sections



Absorption

How much harder are broadband problems?

Bandwidth Response



Metasurfaces

Is perfect cloaking possible over a small bandwidth?

Cloaking

[a]Chao, P. et al. *Maximum electromagnetic local density of states via material structuring*
[b]Amaolo, A. et al. *Physical Limits on Raman Scattering: The Critical Role of Pump and Signal Co-design*
[h]Amaolo, A. and Chao, P., et al. *Maximum Shannon Capacity of Photonic Structures*
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What do limits tell us?

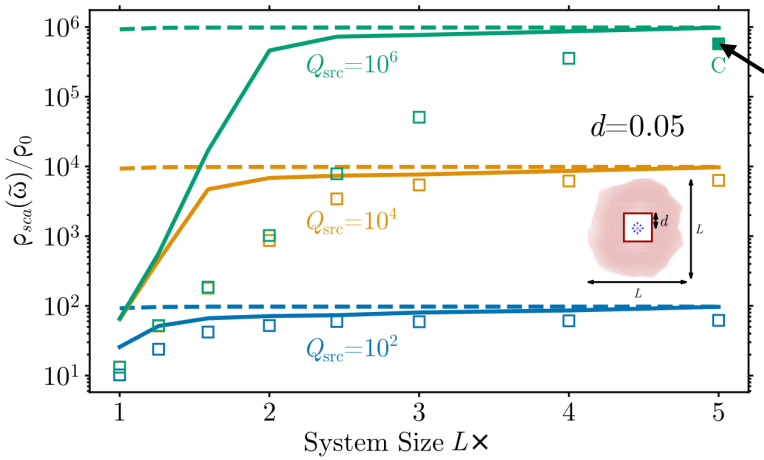
Typically, we know materials, but don't know the structure, so geometry-agnostic & material-dependent limits are especially useful for guiding design.

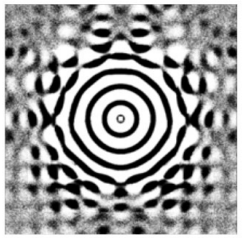
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1. serve as targets for design (certificates of optimality)
2. Inform scaling behavior (with size, material, etc.)
3. Seed inverse design for optimization
4. Inform design via new physics

What do limits tell us?

serve as targets for design (certificates of optimality)

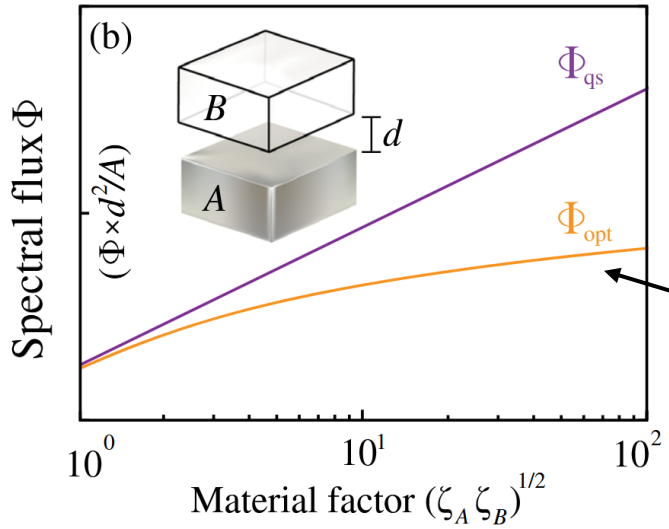




Optimal!

Chao, P. et al. *Maximum electromagnetic local density of states via material structuring*

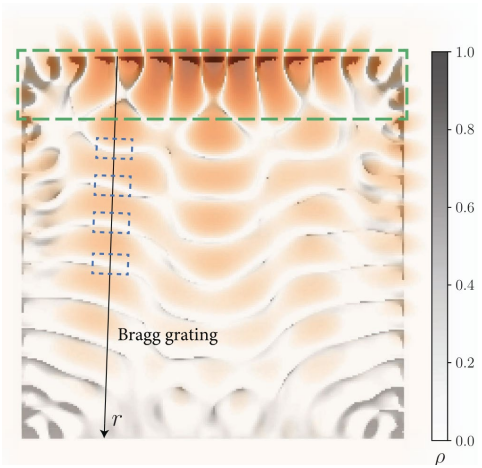
Inform scaling behavior (with size, material, etc.)



Venkataram, P.S., et al. *Fundamental Limits to Radiative Heat Transfer: The Limited Role of Nanostructuring in the Near-Field*

Logarithmic^2 scaling

Seed inverse design for optimization

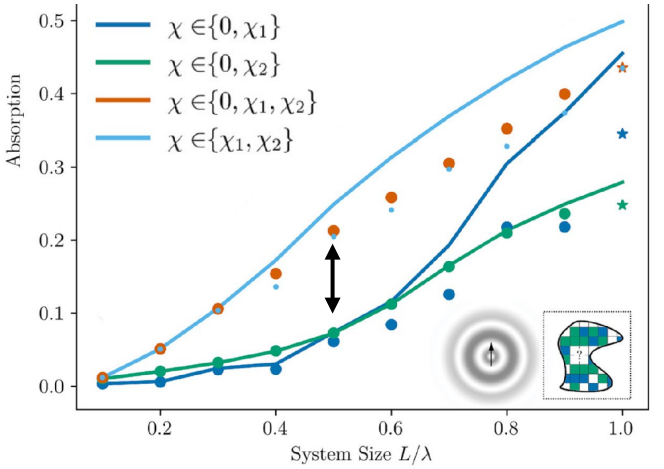


Chao, P.*, Amaolo, A.*, Molesky, S., Rodriguez, A. W. Bounds as blueprints: towards optimal and accelerated photonic inverse design

10 × performance, 10 × fewer iterations compared to standard random-initialized TopOpt

Next!

Inform design via new physics (e.g., certificates of suboptimality)



Amaolo, A. et al. *Can photonic heterostructures provably outperform single-material geometries?*

Multiple materials **necessary** for optimal performance

Multi-material structures exceed single material **bounds**

Next!

Try it yourself:
dolphindes.readthedocs.io

Dolphindes



Dolphindes (very loosely, Dual Optimization Limits for PHotonic/PHysical INverse DESign) is a Python package for calculating limits on the performance of photonic devices using dual optimization methods. It can calculate structure-agnostic performance bounds for a wide range of photonic problems. The package works by relaxing the photonic inverse design problem into a field optimization problem, which can then be further relaxed into a convex problem using Lagrange duality.

Tutorials

Here is a collection of notebooks demonstrating the capabilities of Dolphindes.

Limits

Basic LDOS Limits

GCD Method

Planewave Absorption

Polar Coordinates

Solvers

Using inner PMLs in the polar solver

Using mirror symmetry in the polar solver

Using rotational symmetry in the polar solver

Custom Maxwell Solver

Verlan

Verlan Part 1

```
# Setup geometry
geo = geometry.CartesianFDFDGeometry(
    Nx=Nx, Ny=Ny, Npmlx=Npmlx, Npmly=Npmly, dx=dl, dy=dl
)

ldos_problem = photonics.Photonics_TM_FDFD(
    omega=omega, geometry=geo, chi=chi,
    des_mask=design_mask, ji=ji, chi_background=chi_background,
    sparseQCQP=True, A0=None, s0=None, c0=0.0
)
```

```
result = ldos_problem.bound_QCQP('bfgs', init_lags = None, opt_params=None)
```

Feel free to open issues on GitHub for support.
Can be used for design, limits, and design from limits

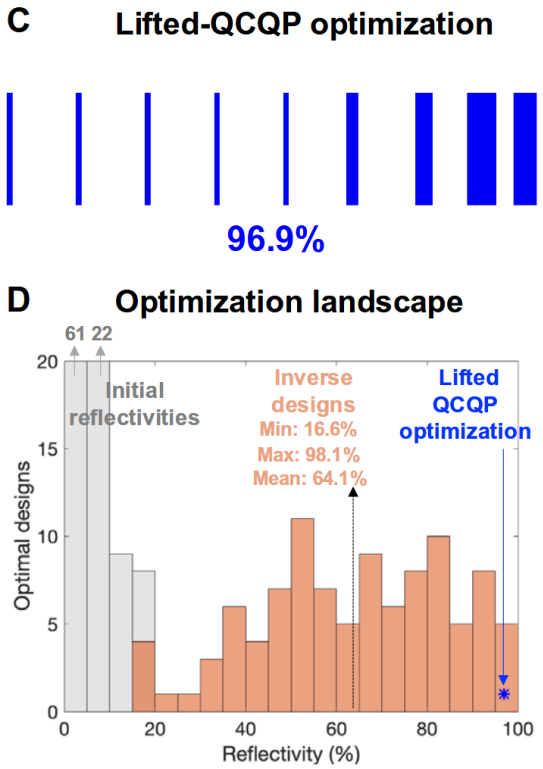
Part 2: Inverse Design from Fundamental Limits

+

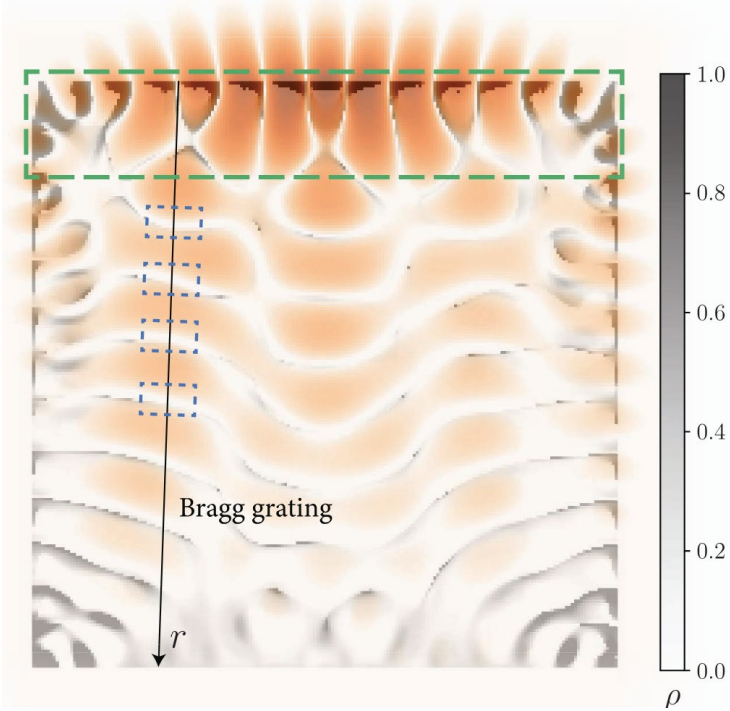
Learning from Suboptimality

Can we exploit knowledge of limits to achieve better designs? **Yes!**

Better performance, fewer iterations, and/or new mechanisms



Gertler, S., Kuang, Z., Christie, C., Li, H., Miller, O. D. *Many photonic design problems are sparse QCQPs*



Chao, P.*, Amaolo, A.*, Molesky, S., Rodriguez, A. W. Bounds as blueprints: towards optimal and accelerated photonic inverse design



$$Q = 1.1 \times 10^{12}$$
$$V = 0.26\,\lambda^2$$

Amaolo, A., Molesky, S., Rodriguez, A.W., Johnson, S.G., Chao, P. The photonic gearbox: ultra-high Purcell enhancement via symmetry breaking. In preparation.

Tutorials

Here is a collection of notebooks

Limits

Basic LDOS Limits

GCD Method

Planewave Absorption

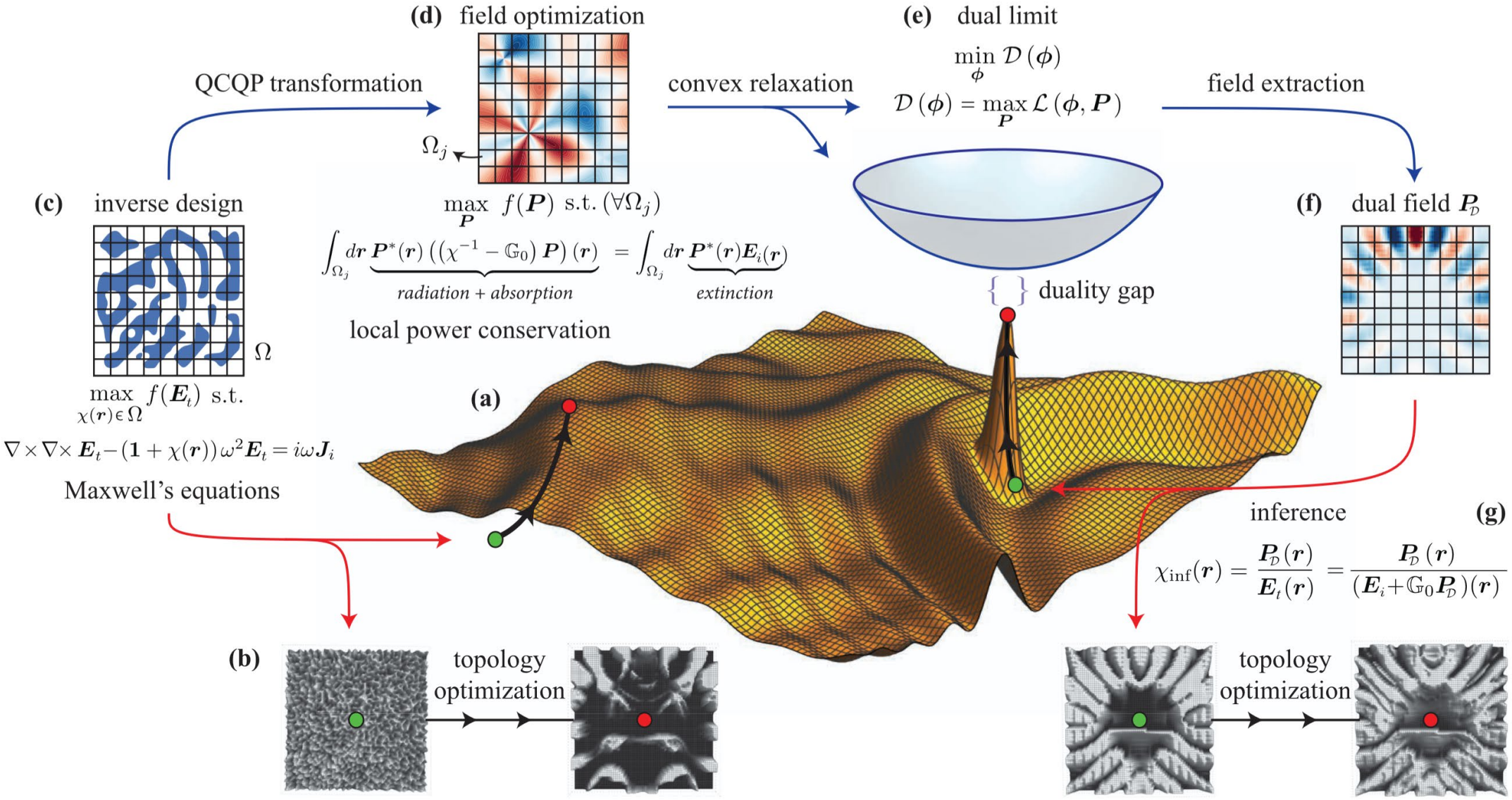
Polar Coordinates

Verlan

Verlan Part 1

Part 2

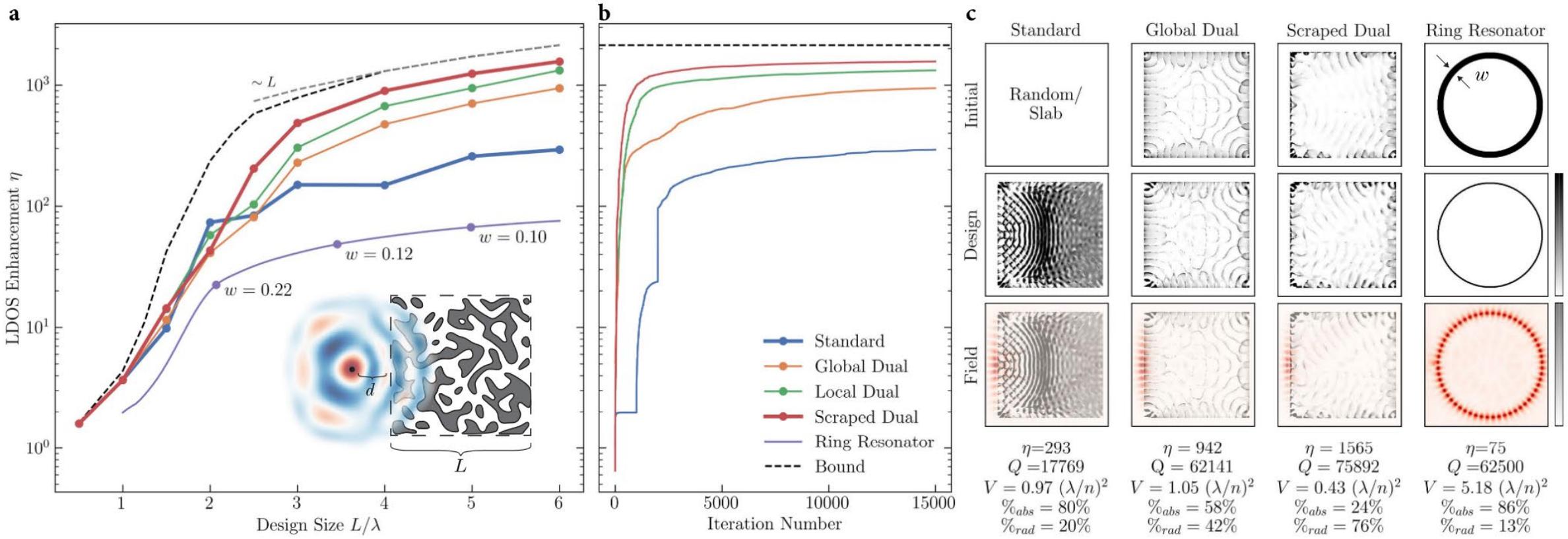
Can we exploit knowledge of limits to achieve better designs?



Simple duality inference has been quite successful in some simple problems!

Can we exploit knowledge of limits to achieve better designs?

Simple duality inference has been quite successful in some simple problems!

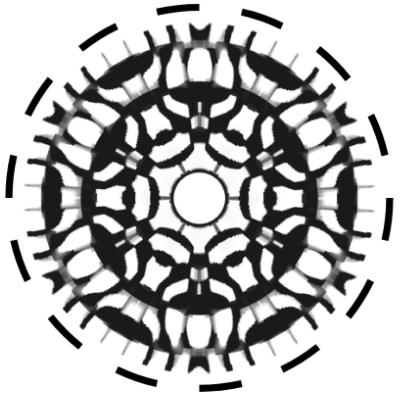


Learning from Suboptimality

LDOS Enhancement and Symmetry Breaking

Setup: enhance single-frequency LDOS by designing a lossless cavity in 2D (TM)

General Structures

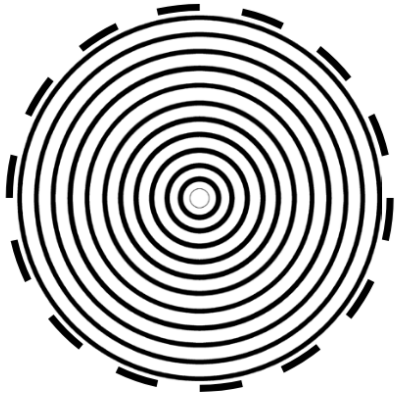


$$\max_{\varepsilon(\boldsymbol{r})} \quad \mathrm{LDOS}(\boldsymbol{r}=0)$$

$$\text{s.t.} \quad \nabla^2 E_z + \omega^2 \varepsilon(\mathbf{r}) E_z = -i\omega\delta(\mathbf{r})$$

Equal?

Cyl. Sym. Structures



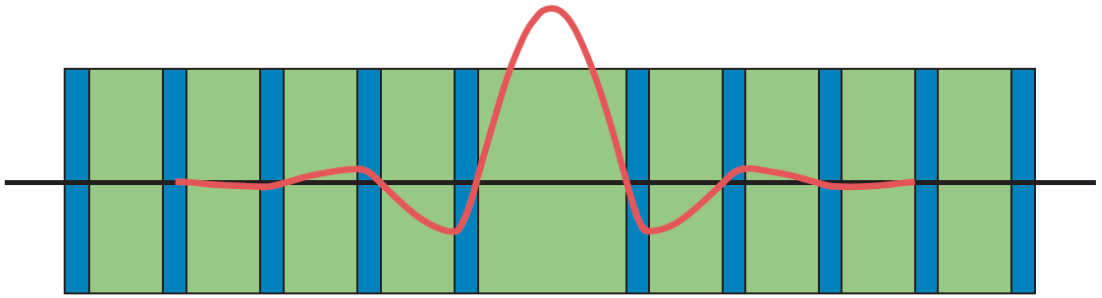
$$\max_{\varepsilon(\boldsymbol{r})} \quad \mathrm{LDOS}(\boldsymbol{r}=0)$$

$$\text{s.t.} \quad \frac{d^2E_z}{dr^2} + \frac{1}{r}\frac{dE_z}{dr} + \omega^2\varepsilon(r)E_z = -\frac{i\omega}{2\pi r}\delta(r)$$

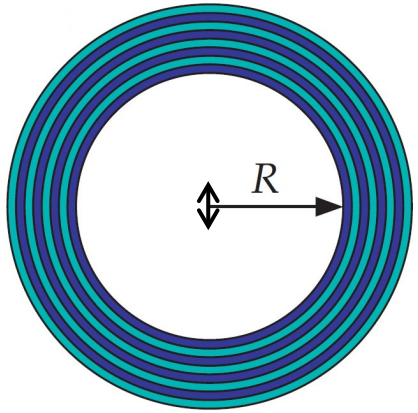
1. We noticed our limits did not give Cyl. Sym. structures…
2. What structures are optimal? Are Bragg onions/fibers optimal?
3. How do we **prove** this?

If the *limit* for Cyl. Sym. Structures is below known general structures, this constitutes a proof of the necessity of symmetry breaking (and we need to find really high-performance structures)

Is the Bragg fiber optimal for LDOS enhancement?



(wrap)



$\text{LDOS} \sim e^{L}$

1D: photonic crystal defect state in bandgap → Huge LDOS
(QWS known to maximize bandgap)

Bragg fiber:

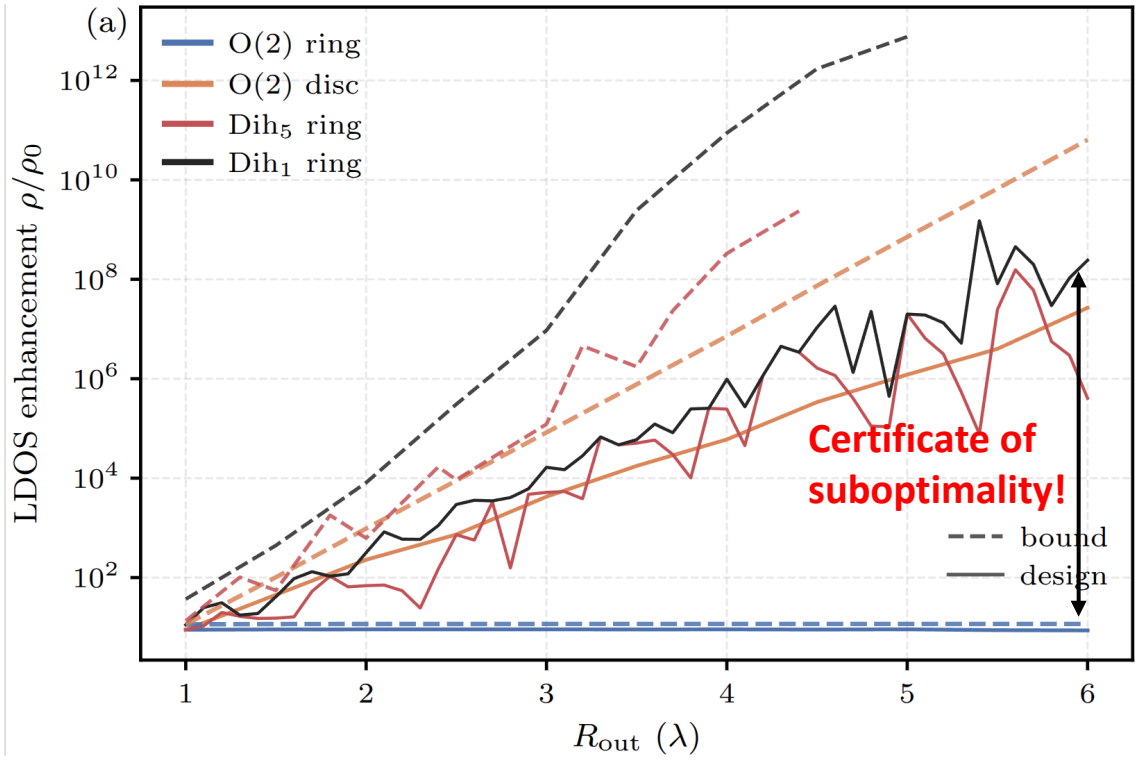
1. Analogous to 1D case which is suspected to be optimal
2. Preserves omnidirectional bandgap?
3. Only couples to $m = 0$ states
4. Perturbatively, $\int \Delta\varepsilon \, |E(r)|^2$ should increase phase mismatch with vacuum

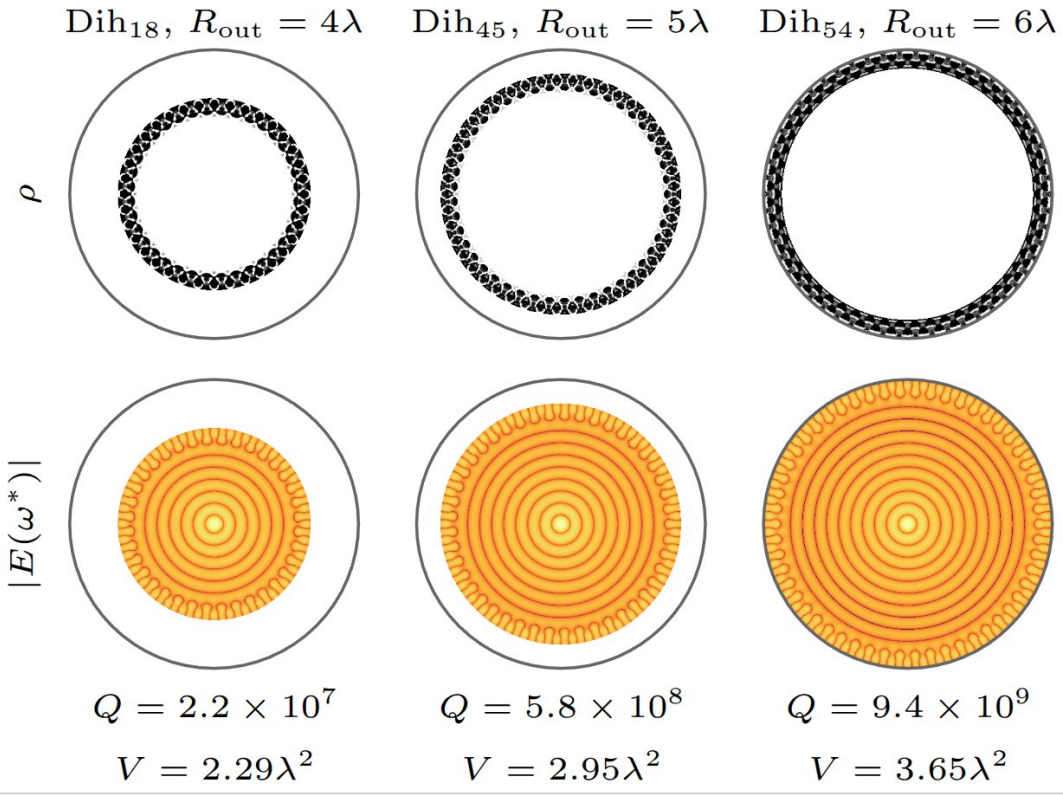
---

But:

1. Fewer degrees of freedom
2. Large gap between structures and limits (are limits loose?)
3. Somehow leverage high-m modes? (WGMs, high-m Bessel waves)

Rings with broken cylindrical symmetry



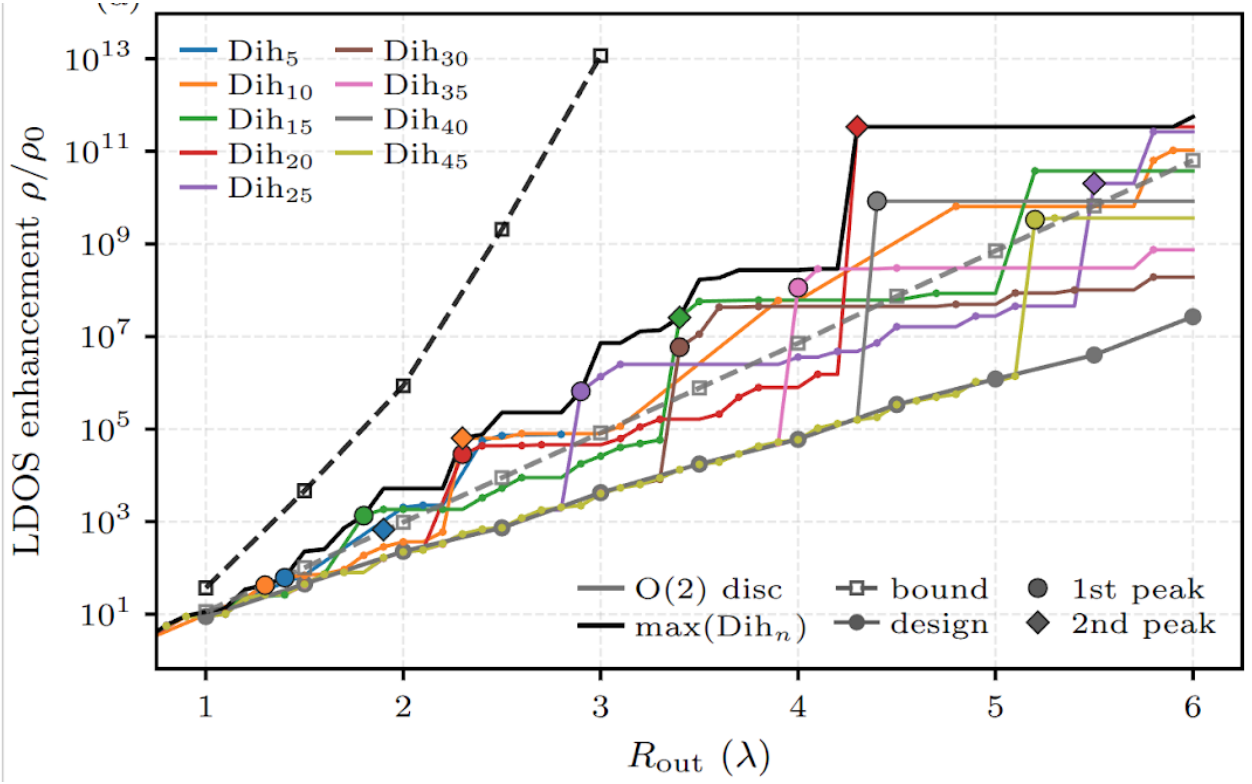


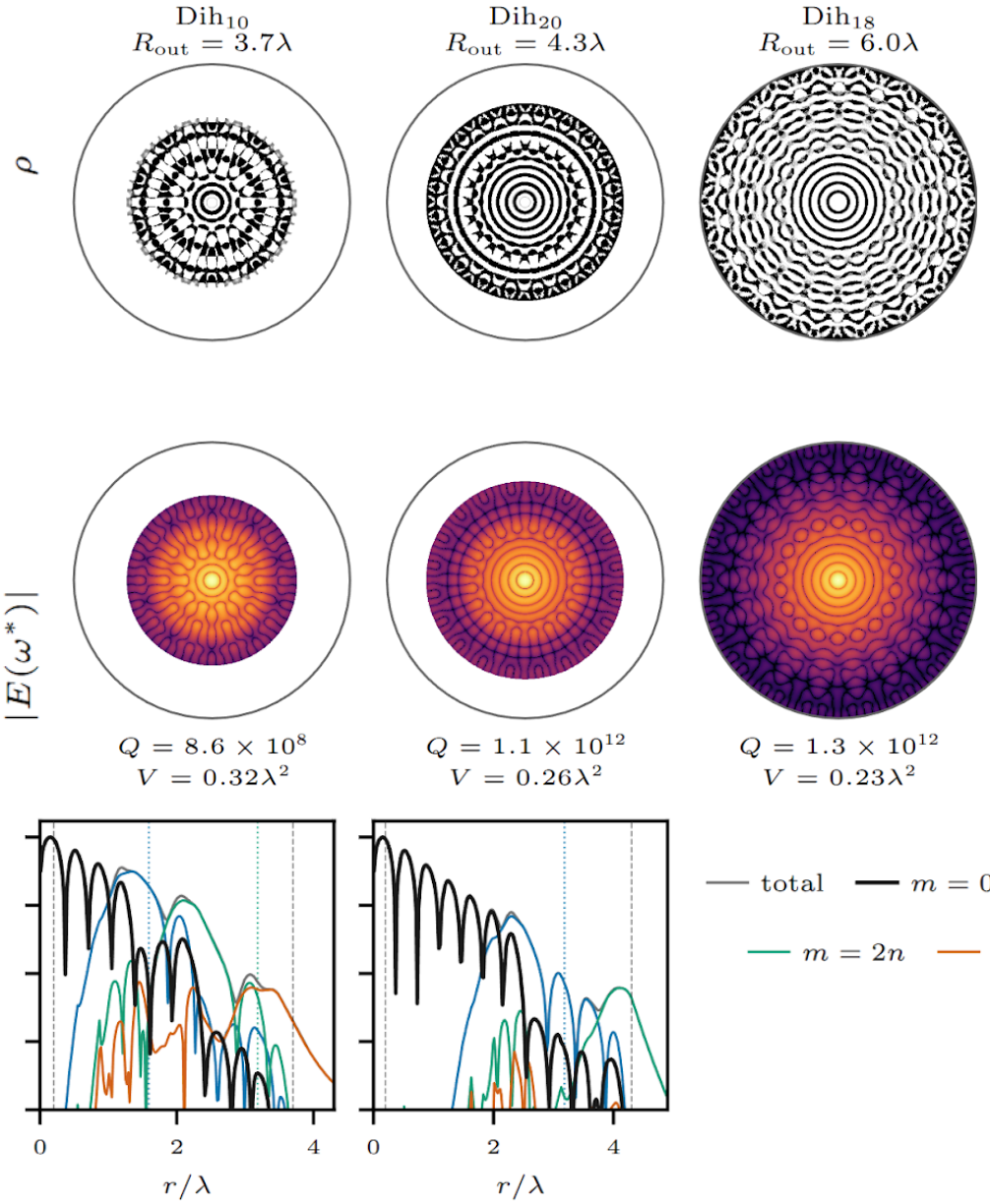
Proves the necessity of symmetry breaking for optimal LDOS enhancement

Large mode volumes but still very high LDOS – favorable for applications compared to sharp tips?

Amaolo, A., Molesky, S., Rodriguez, A.W., Johnson, S.G., Chao, P., *The photonic gearbox: ultra-high local density of states via cylindrical-symmetry breaking.* In preparation.

Gearbox – whole class of new structures



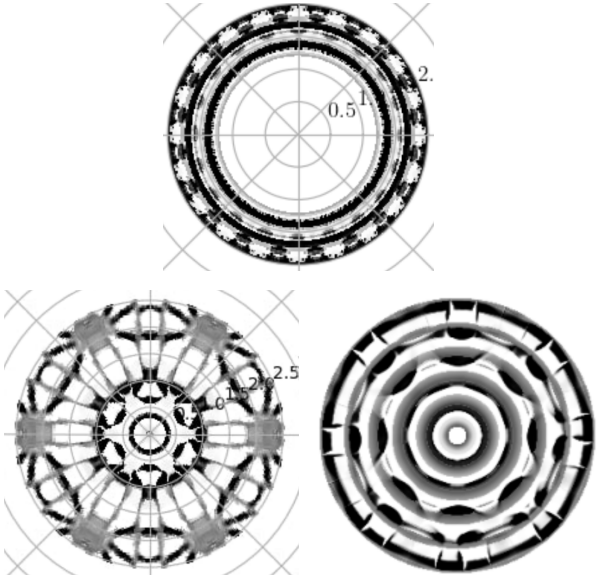


Certificate of suboptimality: structures outperform cylindrically symmetric **bounds**

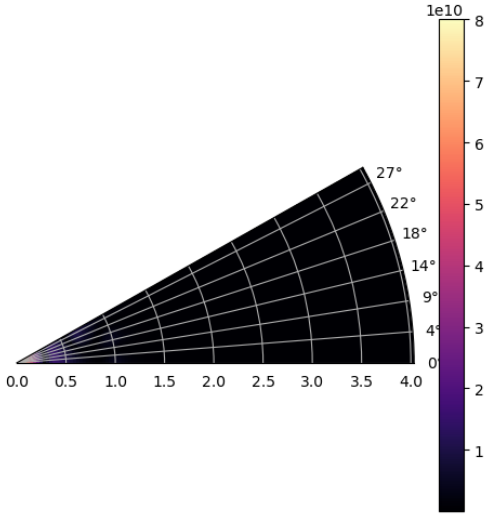
Mechanism (seems to be) **iterative mode conversion**

Only discovered this because bound-inferred polarizations were not symmetric! Only provable because of limits!

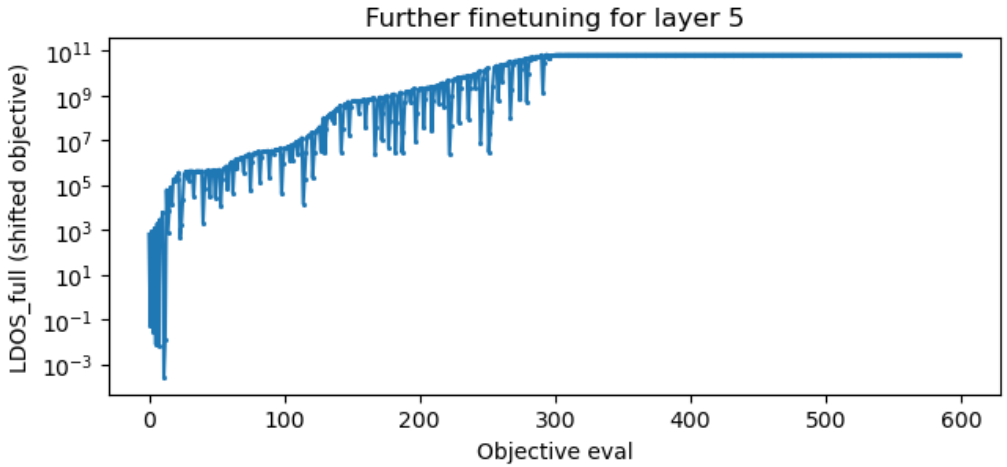
Amaolo, A., Molesky, S., Rodriguez, A.W., Johnson, S.G., Chao, P., *The photonic gearbox: ultra-high local density of states via cylindrical-symmetry breaking.* In preparation.



Class of gearboxes
that perform similarly



Field profiles:
extremely small V



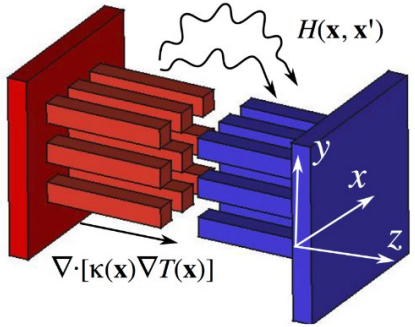
Optimization is
numerically difficult

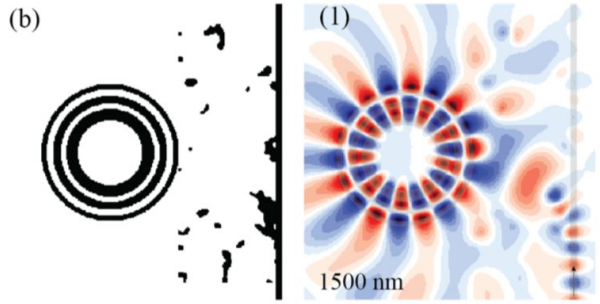
Combining Limits + Inverse Design can:

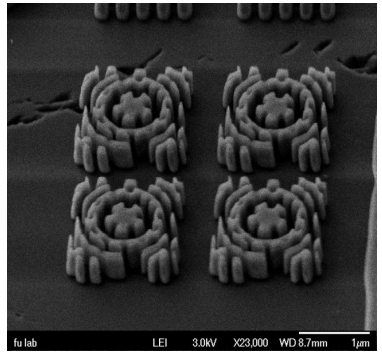
1. Establish **scaling laws** which can in turn inform design
2. Establish **certificates of optimality**
3. Directly be used to **seed optimization**
4. Establish **certificates of *suboptimality*** which can in turn inform design

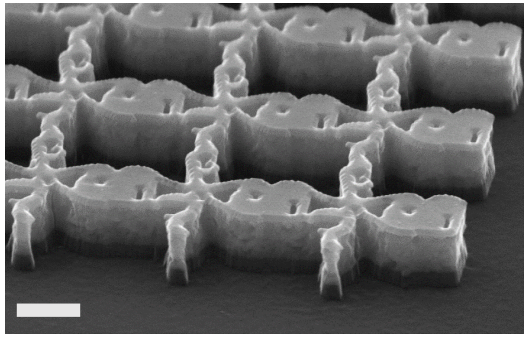
In our group

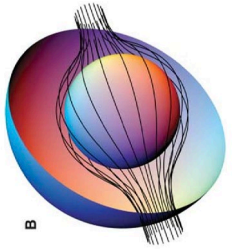
- Large-scale optimization (design)
- Heat transfer (cooling, near field physics)
- Nonlinear optics
- Numerical methods + leveraging ML infrastructure for performance
- Quantum fluctuations (Casimir forces, MEMS, single-photon photodetectors)
- Communication (information theory)
- Wave transformations (signal processing, imaging, cloaking, steering, focusing)

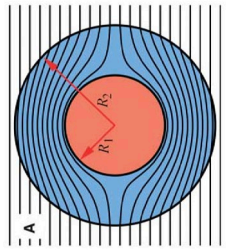




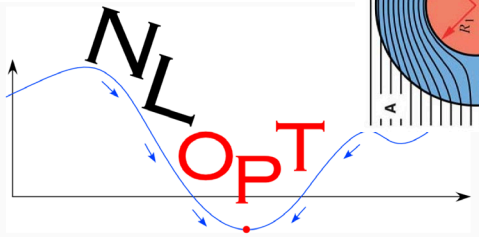












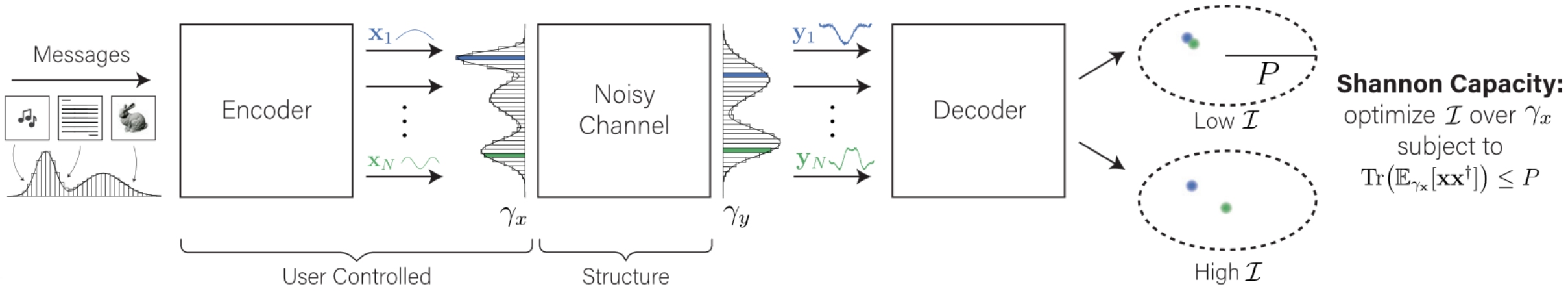


+ physical limits on all these quantities

Part 3: Information Theory of Photonic Communication, Imaging, and Sensing

Information Theory and Shannon Capacity

$$\mathbf{y} = H\mathbf{x} + \mathbf{n} \longrightarrow \mathcal{I}(\mathbf{x};\mathbf{y}) = \int_{\Omega} \gamma_{\mathbf{x}}(\mathbf{x})\, \gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x}) \log \left[\frac{\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})}{\gamma_{\mathbf{y}}(\mathbf{y})} \right] \mathrm{d}\mathbf{y}\mathrm{d}\mathbf{x}$$

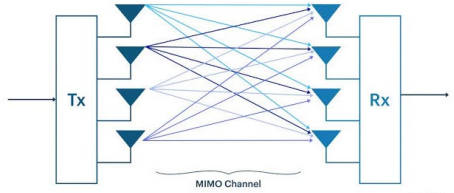


Shannon's insight: C is finite even for error-free transmission!

Information-Theoretic Objectives in Photonics

Communication

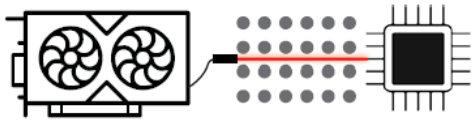
MIMO



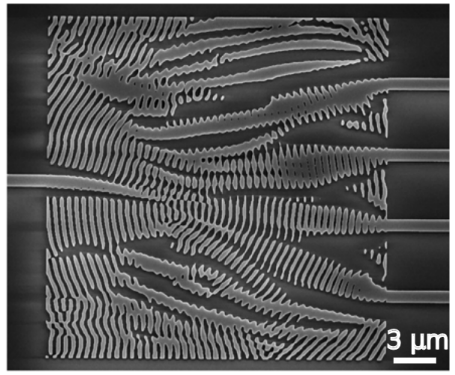
Wireless Transmission



SiPh Communication

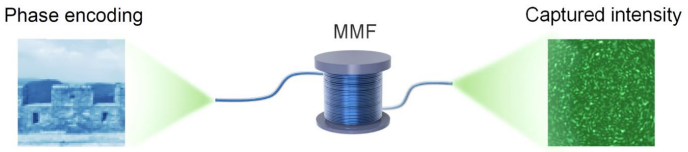


Multiplexing / Demultiplexing

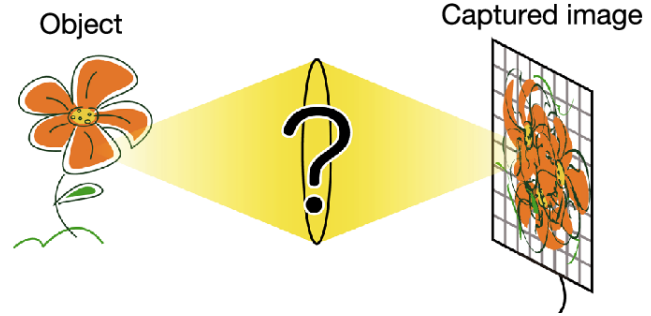


Pita Ruiz, Dalvand, Menard, nat. comm.

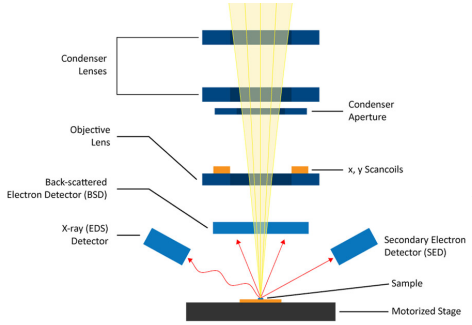
Sensing + Imaging



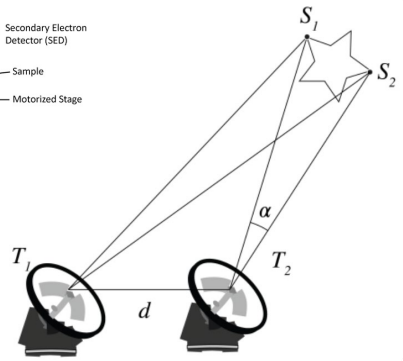
Phase Retrieval + X-ray crystallography



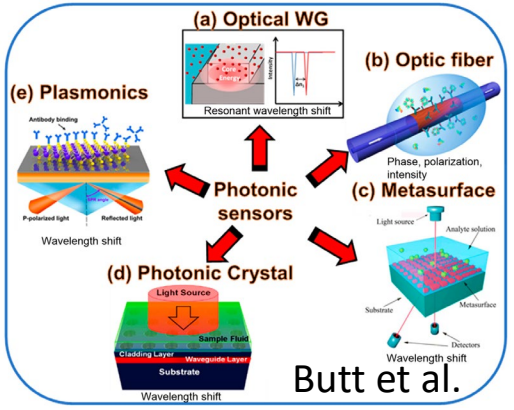
Computational imaging, thermal imaging



Microscopy



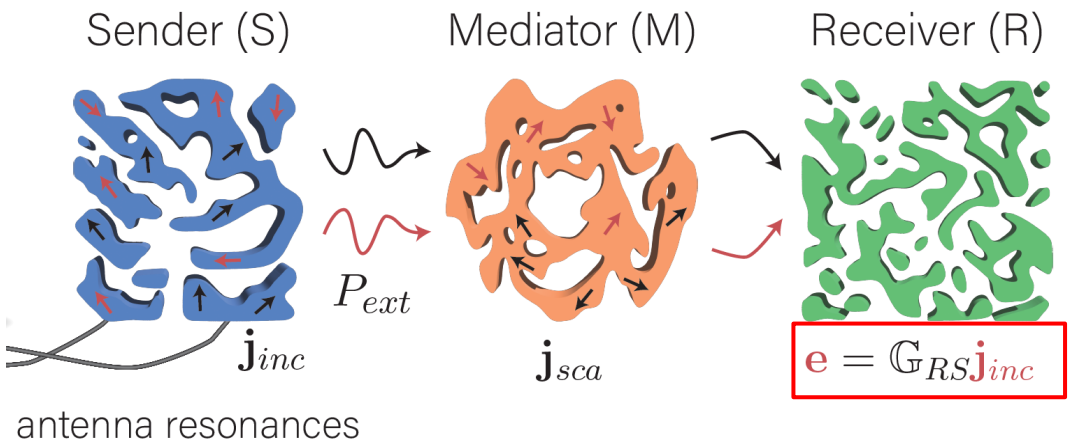
Interferometry



Butt et al.

Photonic Sensing

Intuitive Figures of Merit & Proxy Objectives



$$\mathbb{G}_{RS} = \sum_j \sigma_j \, u_j v_j^{\dagger}$$

Singular vectors are the "modes"
Each mode pair is a "channel"

See work by:
- David Miller
- Owen Miller
- Mats Gustafsson
- Alejandro Rodriguez

$$|e_R|^2, \sum_j \sigma_j^2$$

Signal intensity
Why maximize all channels?

$\mathbb{G}_{RS} \sim$ permutation



Image Resolution
- More information in mixing?
- Exploit statistics/coherence of images?

$$\int \mathrm{Im}\left(\varepsilon(r)\right)|e_R|^2$$

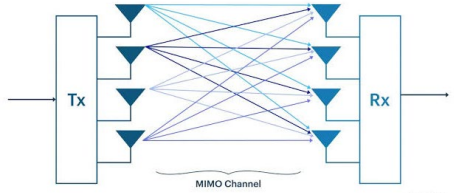
Power Transfer
Not exactly imaging or communication FOM

We should directly be maximizing information-theoretic quantities for communication and sensing

Information-Theoretic Objectives in Photonics

Communication: not just SNR!

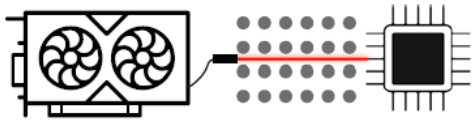
MIMO



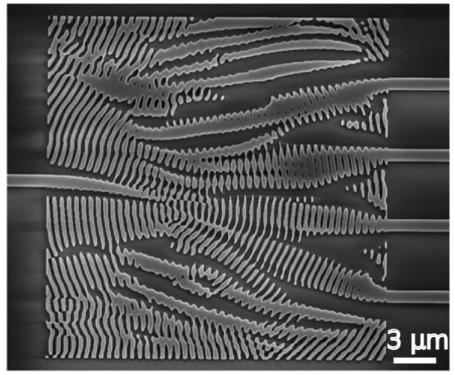
Wireless Transmission



SiPh Communication



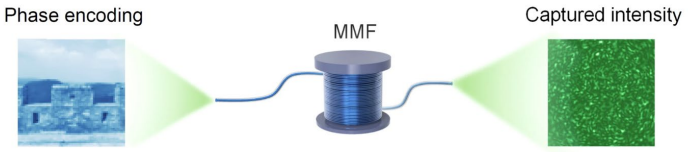
Multiplexing / Demultiplexing



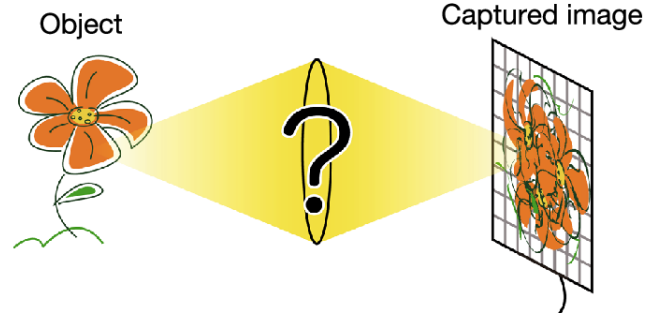
Pita Ruiz, Dalvand, Menard, nat. comm.

How many independent streams can this device carry?

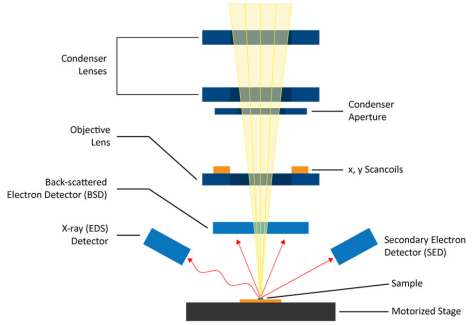
Sensing + Imaging: not just resolution!



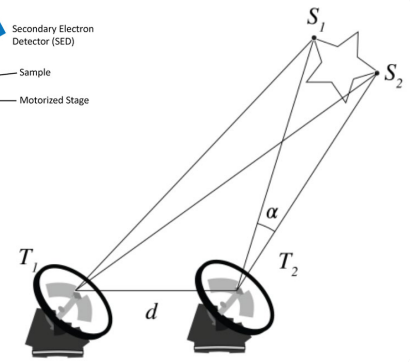
Phase Retrieval + X-ray crystallography



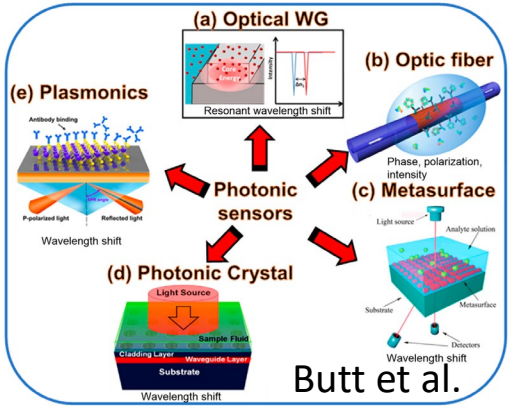
Computational imaging, thermal imaging



Microscopy



Interferometry



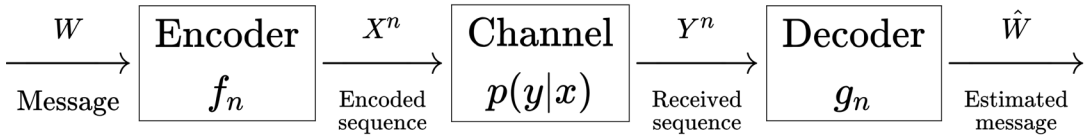
Photonic Sensing

How many distinguishable object states can this device resolve?

i.e., which field patterns can this device keep apart? Resolution and SNR are **proxies** for it, not **definitions** of it

Are photonic channels information channels?

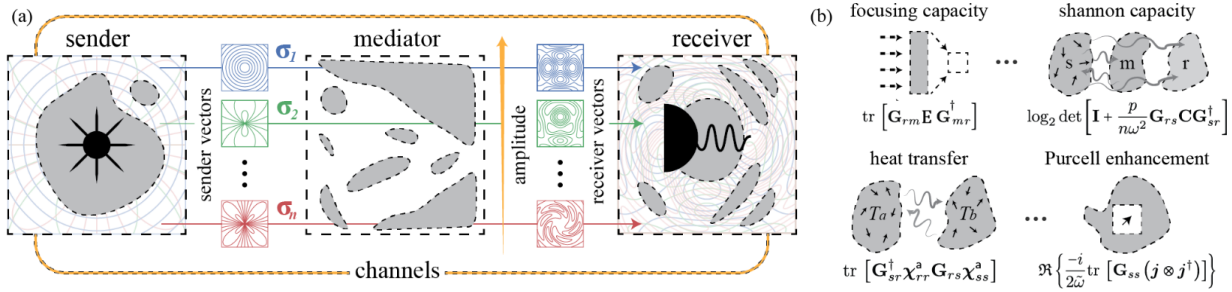
C determines signal propagation



$$C = B\log_2\left(1+\frac{S}{N}\right) \qquad \mathcal{I}(\mathbf{x};\mathbf{y}) = \int_\Omega \gamma_{\mathbf{x}}(\mathbf{x})\,\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})\log\left[\frac{\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})}{\gamma_{\mathbf{y}}(\mathbf{y})}\right]\mathrm{d}\mathbf{y}\mathrm{d}\mathbf{x}$$

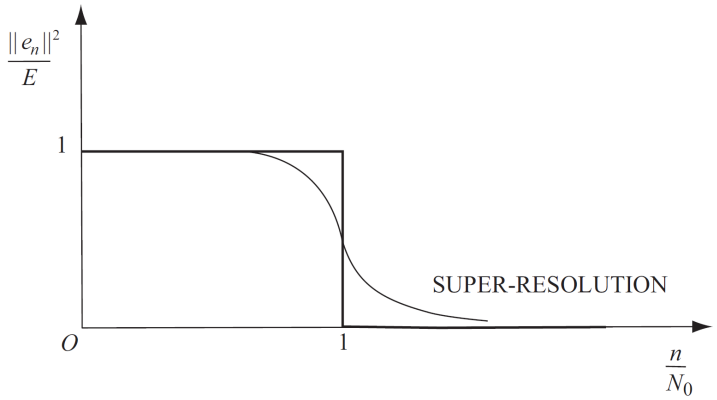
Maximize **mutual information**, entropy loss from reading output signal - natural FOM!

$\mathbb{G}$ determines how the signal propagates

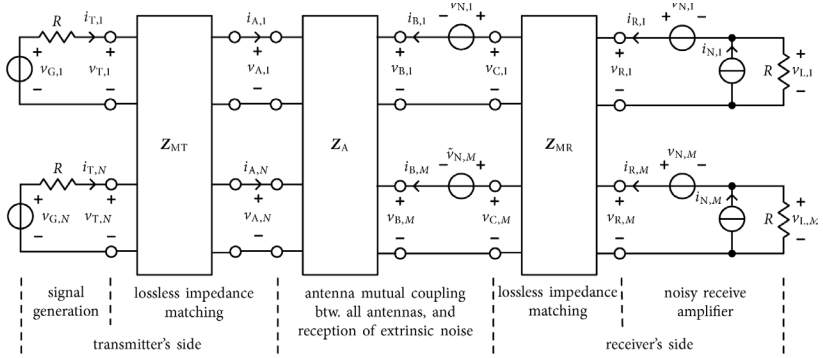


Many important FOM can be written in terms of **channels**

How do we incorporate information theory?



$$\boldsymbol{H} = \begin{bmatrix} e^{-j\frac{2\pi}{\lambda}d_{0,0}} & \cdots & e^{-j\frac{2\pi}{\lambda}d_{0,N_\mathrm{t}-1}} \\ \vdots & \ddots & \vdots \\ e^{-j\frac{2\pi}{\lambda}d_{N_\mathrm{r}-1,0}} & \cdots & e^{-j\frac{2\pi}{\lambda}d_{N_\mathrm{r}-1,N_\mathrm{t}-1}} \end{bmatrix}$$



Number of degrees of freedom

- Poon et al. *Degrees of Freedom in Multiple-Antenna Channels: A Signal Space Approach.* 2005.
- Franceschetti, *Wave Theory of Information.* 2017.
- Miller et al. *Tunnelling escape of waves.* 2025
- Gustafsson, Shadow Area and Degrees of Freedom for Free-Space Communication. 2025.

Far-field approximations

- Brady et al., Interferometric focal planes. 2025.
- Do et al., Reconfigurable ULAs for Line-of-Sight MIMO Transmission

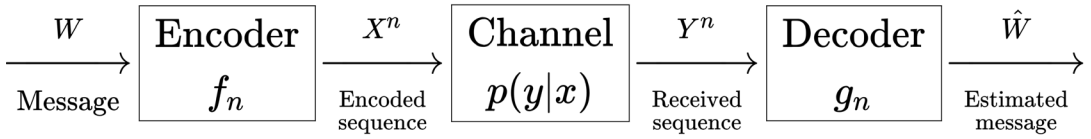
Circuit theory

- Ivrlac and Nossek, *Toward a circuit theory of communication.* 2010.
- Mezghani et al., *Reincorporating Circuit Theory into Information Theory*. 2023.
- Ehrenborg et al., *Capacity Bounds and Degrees of Freedom for*
- *MIMO Antennas Constrained by Q-Factor.* 2025.

& more! David Brady, Mats Gustafsson, David Miller, Owen Miller, Angel Lozano, Massimo Franceschetti, Casimir Ehrenborg, Phillipp del Hougne

Can we explicitly write information-theoretic objectives in terms of inherent quantities of Maxwell's equations?

Recent Example: Shannon Capacity



$$C = B\log_2\left(1+\frac{S}{N}\right)$$

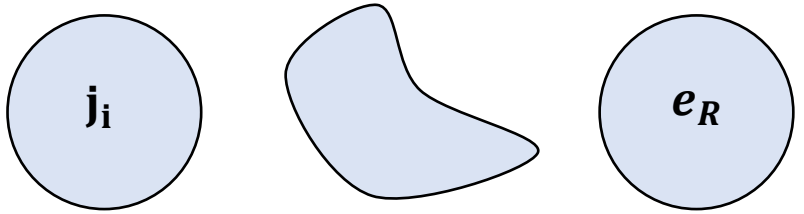
$$\mathcal{I}(\mathbf{x};\mathbf{y}) = \int_{\Omega} \gamma_{\mathbf{x}}(\mathbf{x})\,\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})\log\left[\frac{\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})}{\gamma_{\mathbf{y}}(\mathbf{y})}\right]\mathrm{d}\mathbf{y}\mathrm{d}\mathbf{x}$$

$$y = Hx + n$$

We optimize the distribution of sender x so we can learn the most about it by reading y

$$C = \log\det\left(I + \frac{1}{N}\mathrm{H}QH^{\dagger}\right)$$

$$Q = \mathbb{E}\left[xx^{\dagger}\right]$$



Sender (S) | Mediator (M) | Receiver (R)

$$e_R = \mathbb{G}_{RS}j_i + n$$

We optimize the distribution of sender j_i so we can learn the most about it by reading e_R

$$C(\epsilon, \mathbb{J}_i) = \log_2\det\left(\mathbb{I} + \frac{P}{N\omega^2}\mathbb{G}_{RS}(\epsilon)\mathbb{J}_i\mathbb{G}_{RS}^{\dagger}(\epsilon)\right)$$

$$\mathbb{J}_i = \mathbb{E}\left[j_i j_i^{\dagger}\right]$$

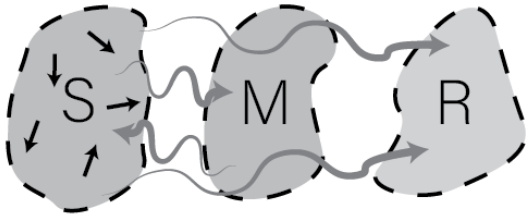
Familiar $\log\det(\cdot)$ expression

$$\sum_{i=1}^{n_c}\log_2(1+\gamma J_i|\sigma_i|^2)$$

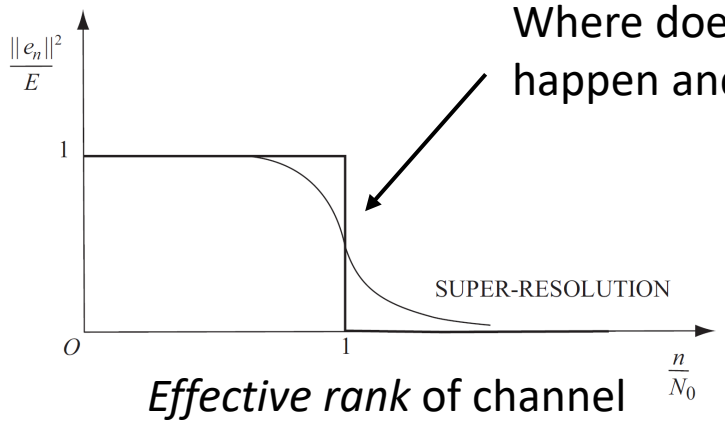
Amaolo, A., Chao, P., Strekha, B., Clarke, S., Mohajan, J., Chen, F.C., Molesky, S., Rodriguez, A.W. *Maximum Shannon Capacity of Photonic Structures*

The **spectrum** and **decay** singular values is key. Example:

Shannon Capacity



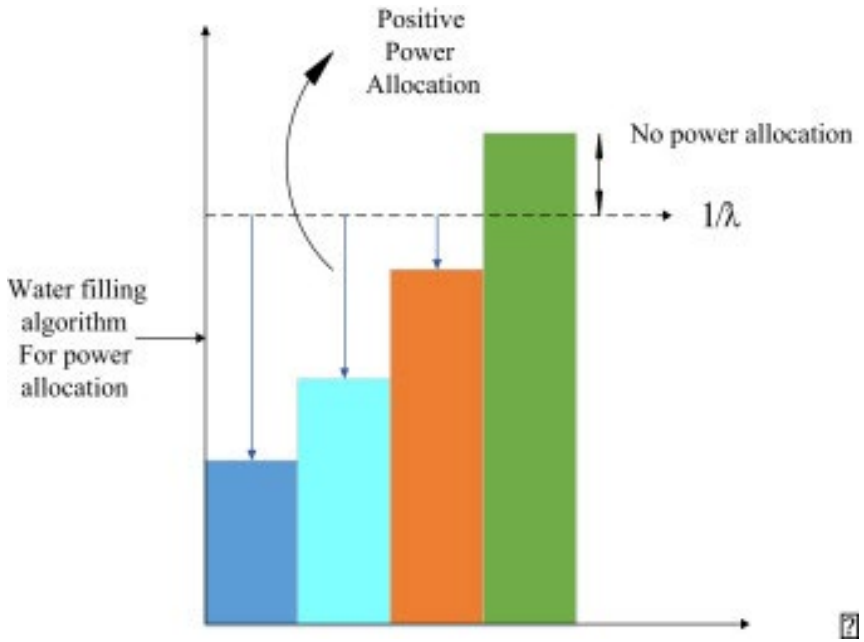
$$\sum_i \log_2\left(1 + j_i \sigma_i^2(\mathbb{G}_{RS})\right)$$



Where does this happen and why?

Effective rank of channel

Water-filling: given a spectrum, how do you allocate power?

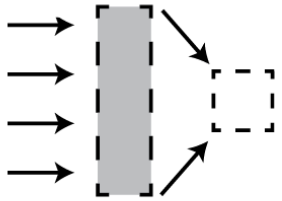


Some channels remain unused

We seek to understand **what spectra are allowable** under Maxwell's equations (i.e., fundamental limits) and how to **engineer these limits**

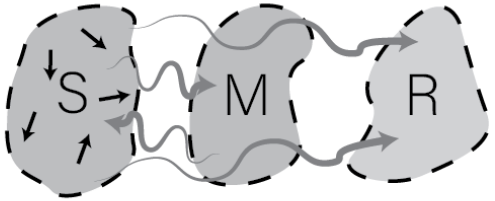
Photonic channels directly relate to information-theoretic quantities

Focusing Capacity



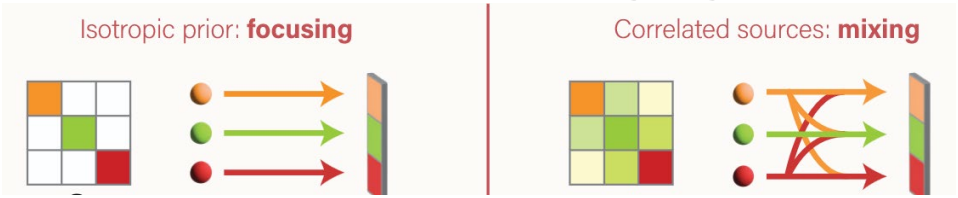
$$\sum_i \sigma_i^2(\mathbf{G}_{MR})$$

Shannon Capacity



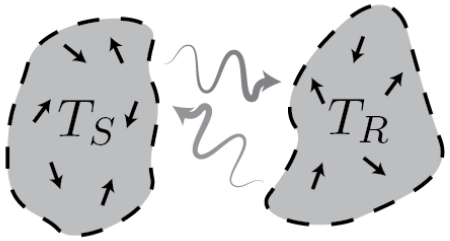
$$\sum_i \log_2\left(1+\mathrm{SNR}\sigma_i^2(\mathbf{G}_{RS})\right)$$

Incoherent Imaging



$$\frac{1}{2}\sum \log_2\left(1+\mathrm{SNR}\sigma_i^2(\mathbf{G}_{RS}^* \odot \mathbf{G}_{RS})\right)$$

Heat Transfer



$$\frac{2}{\pi}\frac{\mathrm{Im}\chi_S}{|\chi_S|^2}\sum_i \sigma_i(\mathrm{Proj\ Asym}\, G_{RS})\mathbf{p}_j^\dagger \mathbf{p}_j)$$

Purcell Enhancement



$$\frac{2\omega}{\pi}\sum_i \sigma_i^2(\mathrm{Asym}\, \mathbf{G}_{SS})$$

Phase Retrieval



$$\sum \log_2\left(1+\mathrm{SNR}\sigma_i^2 \mathrm{vec}(\mathbf{G}_{RS}^\dagger \otimes \mathbf{G}_{RS})\right)$$

Enough to do inverse design and **co-optimize** structure with information theory!

More fundamental question: **what spectra are possible?**

- Amaolo, A., Chao, P., Strekha, B., Clarke, S., Mohajan, J., Chen, F.C., Molesky, S., Rodriguez, A.W. *Maximum Shannon Capacity of Photonic Structures*
- Virally, P., Chao, P., Amaolo, A., Rodriguez, A.W., Molesky, S., *Indexed singular value bounds on scattering operators: How many channels can a photonic device support?*
- Chen, F.J., Amaolo, A., Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. *Optically Incoherent Photonic Mutual Information*
- Kuang, Z., Miller, O. *Bounds on the Coupling Strengths of Communication Channels and their Information Capacities*

Singular Value Limits

Courant-Fischer-Weyl

$$\sigma_n^2 = \min_{Q_{n-1} \subseteq X} \max_{\boldsymbol{x} \in Q_{n-1}^{\perp}} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x} = \max_{Q_n \subseteq X} \min_{\boldsymbol{x} \in Q_n} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x}$$

$\|x\|^2 = 1$

$$\min_{\boldsymbol{x} \in Q_n} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x} \leq \sigma_n^2 \leq \max_{\boldsymbol{x} \in Q_{n-1}^{\perp}} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x}.$$

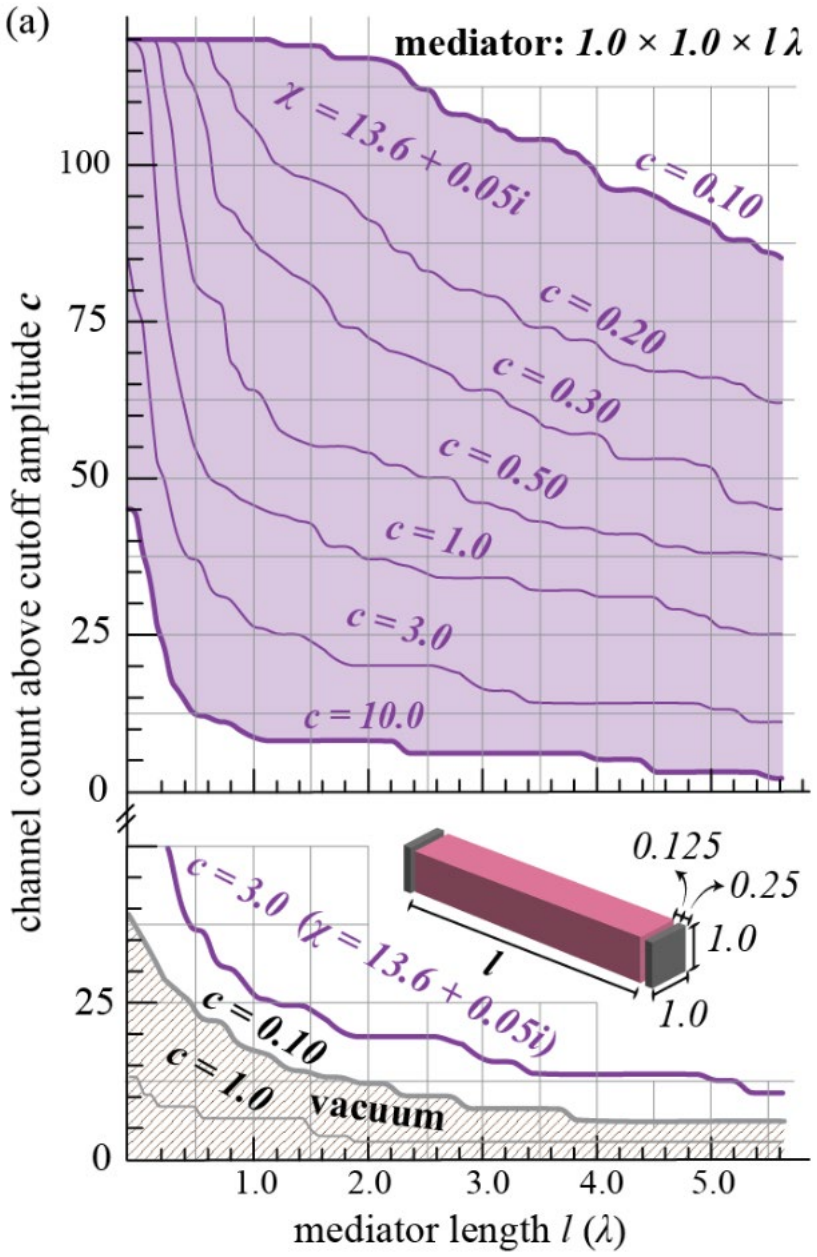
Basically: optimize sources in the correct subspace → quadratic programs amenable to limits above

Paul Virally



Virally, P., Chao, P., Amaolo, A., Rodriguez, A.W., Molesky, S., *Indexed singular value bounds on scattering operators: How many channels can a photonic device support?*

Singular Value Limits



Effective rank of channel

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$$\|x\|^2 = 1$$

$$\min_{\boldsymbol{x}\in Q_n} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x} \leq \sigma_n^2 \leq \max_{\boldsymbol{x}\in Q_{n-1}^{\perp}} \boldsymbol{x}^\dagger \mathbf{A}^\dagger \mathbf{A} \boldsymbol{x}.$$

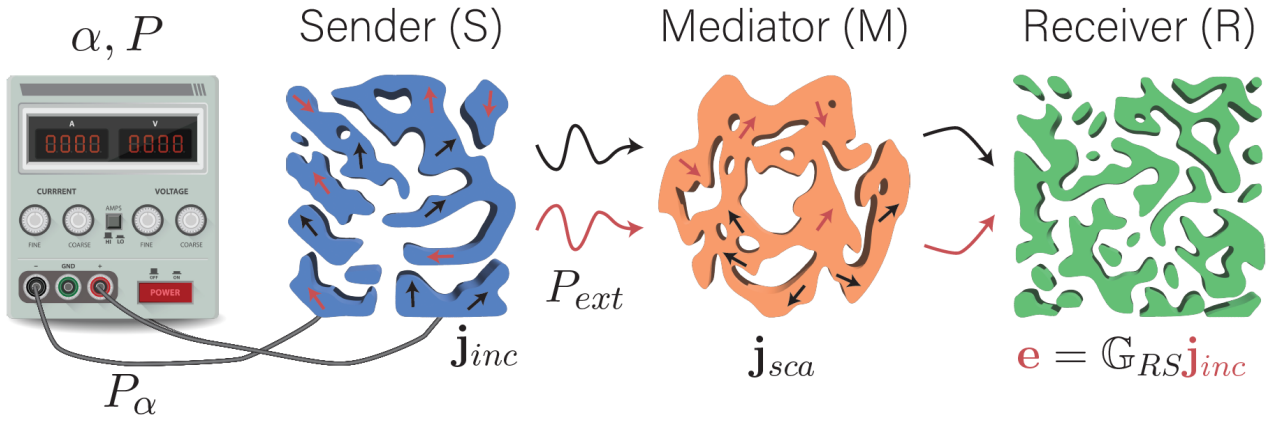
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Virally, P., Chao, P., Amaolo, A., Rodriguez, A.W., Molesky, S., *Indexed singular value bounds on scattering operators: How many channels can a photonic device support?*

Shannon Capacity: recently well-understood



Coherent communication problem:

- Control the distribution of sources
- Sources are coherent (have phase relationships)
- Receivers are coherent (can measure phase)

$$\max_{\mathbf{J},|\boldsymbol{\sigma}|^2} \quad \sum_{i=1}^{n_c} \log_2(1+\gamma J_i|\sigma_i|^2)$$

$$s.t. \quad J_i \geq 0,\ |\sigma_i|^2 \geq 0,\ \forall i \leq n_c$$

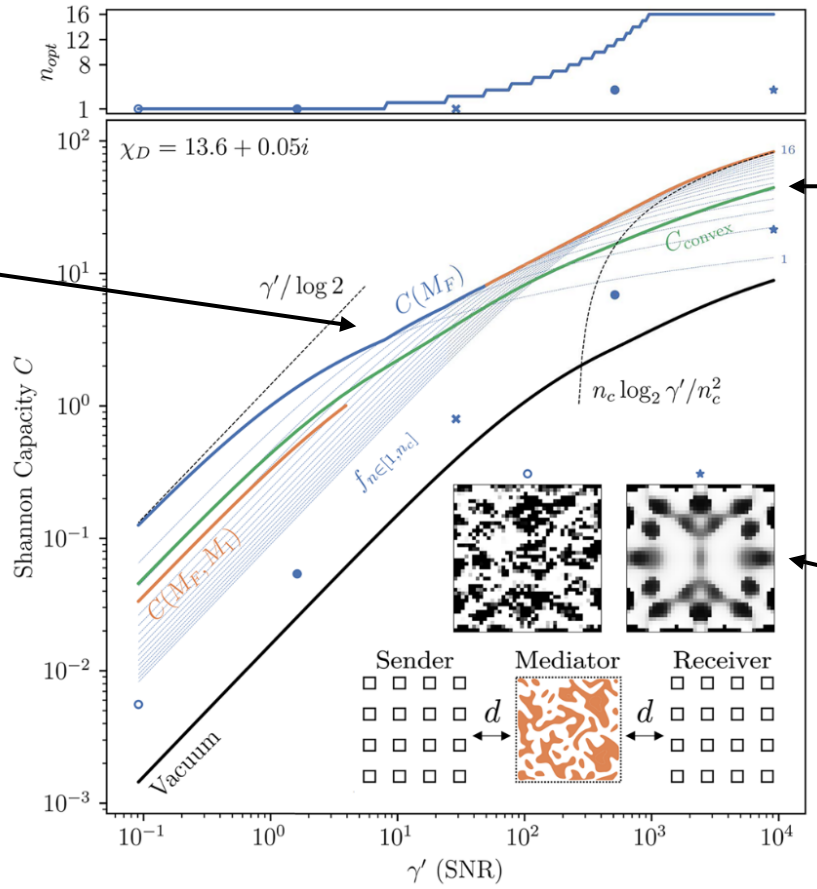
$$\sum_i J_i \leq 1$$

→ Limits on Shannon capacity over **sender distributions** (signals) and **photonic structuring**

Amaolo, A., Chao, P., Strekha, B., Clarke, S., Mohajan, J., Chen, F.C., Molesky, S., Rodriguez, A.W. *Maximum Shannon Capacity of Photonic Structures*
See also: Kuang, Z., Miller, O. *Bounds on the Coupling Strengths of Communication Channels and their Information Capacities*

Limits on Shannon capacity subject to $\sum_i \sigma_i^2 < M_F$

Limits reproduce expected channel count scaling



Within one order of magnitude of inverse design

Co-optimized power allocation with photonic structuring

Bounds on σ_i may give tighter limits!

- What if the **receiver can't measure phase?**
- What if **sources are not coherent?**
- What if we **can't control the source distribution,** but sources are **still correlated?**

More realistic conditions for **sensing**

Amaolo, A., Chao, P., Strekha, B., Clarke, S., Mohajan, J., Chen, F.C., Molesky, S., Rodriguez, A.W. *Maximum Shannon Capacity of Photonic Structures*

Mutual information of sensing + imaging

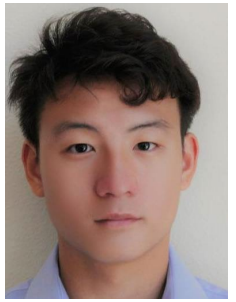
	Coherent Receiver	Incoherent Receiver What if the receiver can't measure phase?
Coherent Sources	$e_R = \mathbb{G}_{RS} j_i + n$ $C = \log_2 \det\left(\mathbb{I} + \frac{1}{N\omega^2}\mathbb{G}_{RS}\mathrm{Q}\mathbb{G}_{RS}^{\dagger}\right)$ Shannon Capacity	$e_R = \lvert\mathbb{G}_{RS} j_i + n\rvert^{\odot 2}$ $\mathcal{I}(x;y) = \int\int \gamma(x)\,\gamma(y\lvert x)\,\log\left[\frac{\gamma(y\lvert x)}{\gamma(y)}\right]\mathrm{d}y\mathrm{d}x$ Phase retrieval Not logdet, hard! Still computable
Incoherent Sources What if sources are not coherent?	Still in progress ☺	$\langle \mathrm{I}_R \rangle = (\mathbb{G}^*_{RS} \odot \mathbb{G}_{RS})\mathrm{B} + n_{\mathrm{I}}$ $\quad B = \langle \lvert j_i\rvert^{\odot 2} \rangle$ $I = \frac{1}{2}\log_2 \det\left(\mathbb{I} + \frac{1}{N}(\mathbb{G}_{RS}{}^* \odot \mathbb{G}_{RS})\mathbf{Q}_{\mathrm{B}}(\mathbb{G}_{RS}{}^* \odot \mathbb{G}_{RS})^{\dagger}\right)$ Intensity MIMO, conventional imaging

- Receiver and source coherence are properties of the channel model
- Source covariance Q is either controlled (communication) or not controlled (imaging, sensing)

Shannon capacity

Mutual information

Francis Chen



Chen, F.J., Amaolo, A., Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. *Optically Incoherent Photonic Mutual Information*

Mutual information of Phase Retrieval

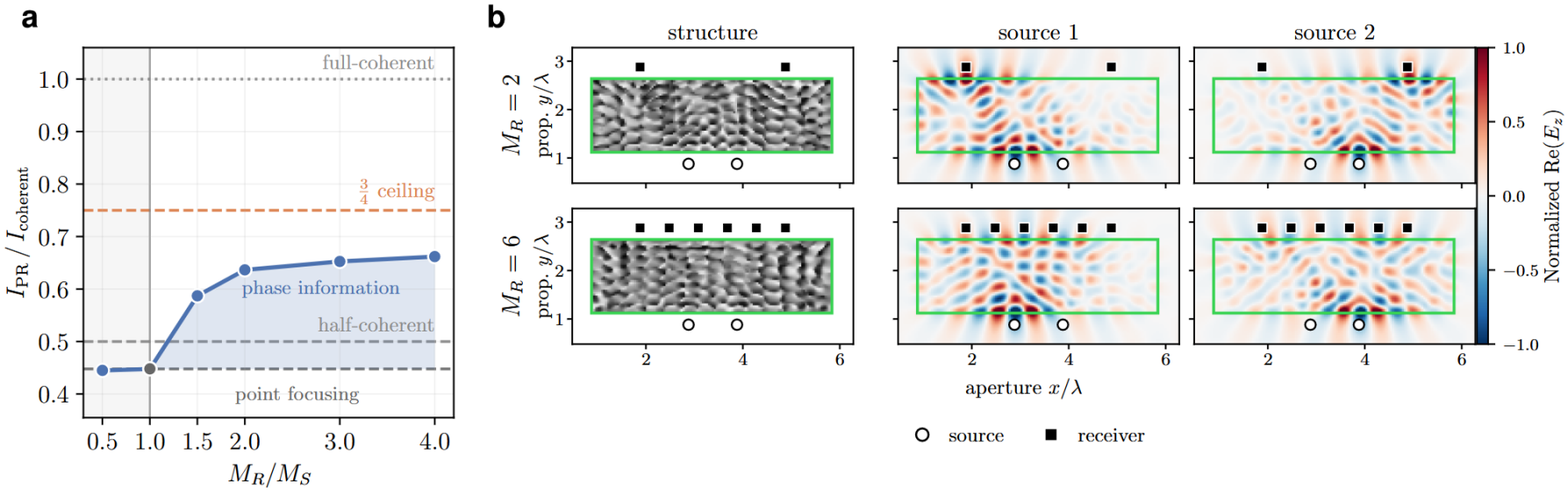
$$\mathrm{y} = |\mathrm{Hx} + n|^2 \longrightarrow \mathcal{I}(\mathbf{x};\mathbf{y}) = \int_\Omega \gamma_{\mathbf{x}}(\mathbf{x})\,\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x}) \log\left[\frac{\gamma_{\mathbf{y}|\mathbf{x}}(\mathbf{y}|\mathbf{x})}{\gamma_{\mathbf{y}}(\mathbf{y})}\right] \mathrm{d}\mathbf{y}\mathrm{d}\mathbf{x}, \longrightarrow$$

Heteroscedastic noise

Inverse design via Monte Carlo integration

Fix $j_i \sim \mathcal{N}(0, Q)$

$$\mathcal{J}(\mathbf{x};\mathbf{y}) = \underbrace{\log\det\left(\mathbb{I}_{M_R} + \frac{1}{N}\mathbb{G}_{t,RS}\mathbf{Q}\mathbb{G}_{t,RS}^\dagger\right)}_{\text{Coherent value}} + \underbrace{\int_{\mathbb{C}^{M_S}}\int_{\mathbb{R}_{>0}^{M_R}} \gamma_{\mathbf{x}}\,\gamma_{\mathbf{y}|\mathbf{x}}\left(\log\left[\frac{g_2(\mathbf{y}|\mathbf{x})}{g_1(\mathbf{y})}\right]\right) d\mathbf{y}\,d\mathbf{x}}_{\text{Negative correction}}, \xrightarrow{\text{High-SNR}} \frac{1}{2}\log\det(\cdot) + o(1) - 0.48$$



Not yet answered: what is possible in terms of channel engineering?

Even with flat priors, we can design a photonic structure that mixes sources to gain information

$$g_2(\mathbf{y}|\mathbf{x}) \equiv N^{M_R}\gamma_{\mathbf{y}|\mathbf{x}}$$

$$g_1(\mathbf{y}) \equiv \det\left(N\mathbb{I}_{M_R} + \mathbb{G}_{t,RS}\mathbf{Q}\mathbb{G}_{t,RS}^\dagger\right)\gamma_{\mathbf{y}}$$

Chen, F.J., Amaolo, A., Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. *Optically Incoherent Photonic Mutual Information*

Mutual information of sensing + imaging

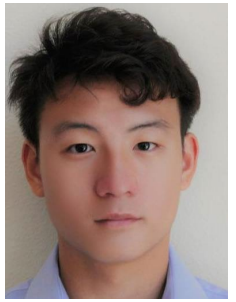
	Coherent Receiver	Incoherent Receiver What if the receiver can't measure phase?
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Incoherent Sources What if sources are not coherent?	Still in progress ☺	$\langle \mathrm{I}_R \rangle = (\mathbb{G}^*_{RS} \odot \mathbb{G}_{RS})\mathrm{B} + n_{\mathrm{I}}$ $\quad B = \langle \lvert j_i\rvert^{\odot 2} \rangle$ $I = \frac{1}{2}\log_2 \det\left(\mathbb{I} + \frac{1}{N}(\mathbb{G}_{RS}{}^* \odot \mathbb{G}_{RS})\mathbf{Q}_{\mathrm{B}}(\mathbb{G}_{RS}{}^* \odot \mathbb{G}_{RS})^{\dagger}\right)$ Intensity MIMO, conventional imaging

- Receiver and source coherence are properties of the channel model
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Shannon capacity

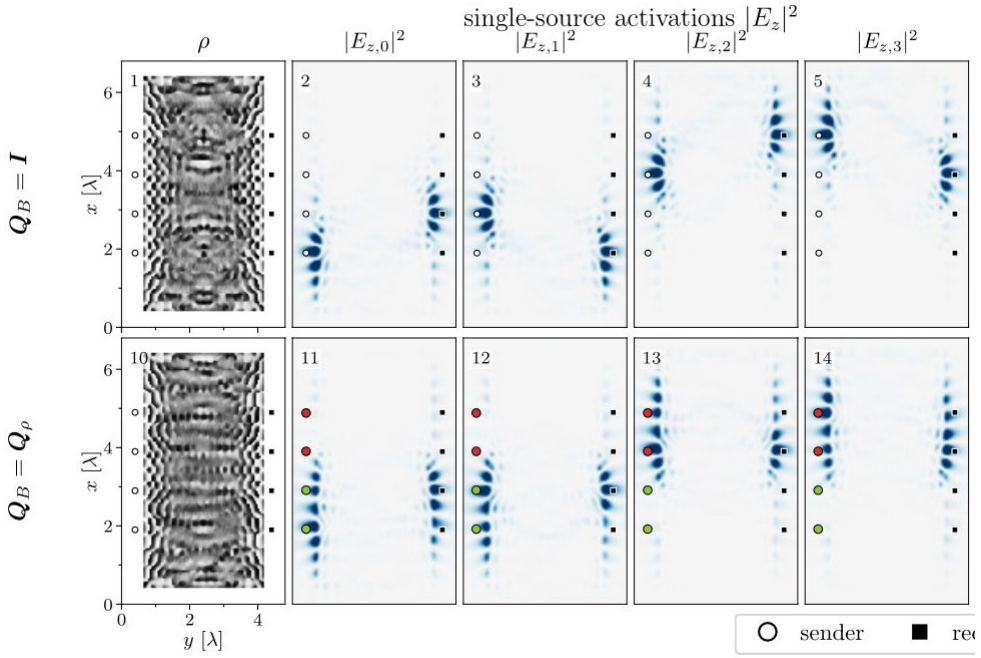
Mutual information

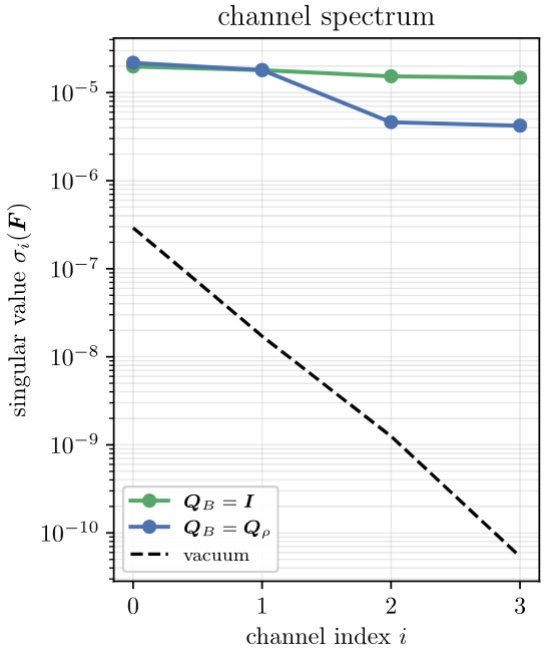
Francis Chen



Chen, F.J., Amaolo, A., Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. *Optically Incoherent Photonic Mutual Information*

Mutual information of incoherent imaging



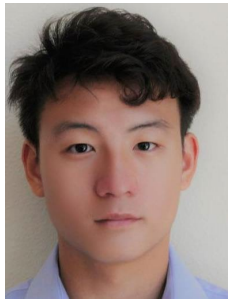


- General-purpose imagers have $\mathbf{Q_B} \propto \mathbb{I}$ → point-focusing optima (when the structure *can* flatten the spectrum).
- Mutually incoherent sources can still have correlations → *flattenability* of spectrum is key

$$\mathcal{I} = \frac{1}{2}\sum_{i=1}^{M} \log\left(1 + \frac{q_B}{N_B}\,\sigma_{\mathbb{F},i}^2\right) \longrightarrow$$

Limits on imaging systems and limits on flattenability (in progress)

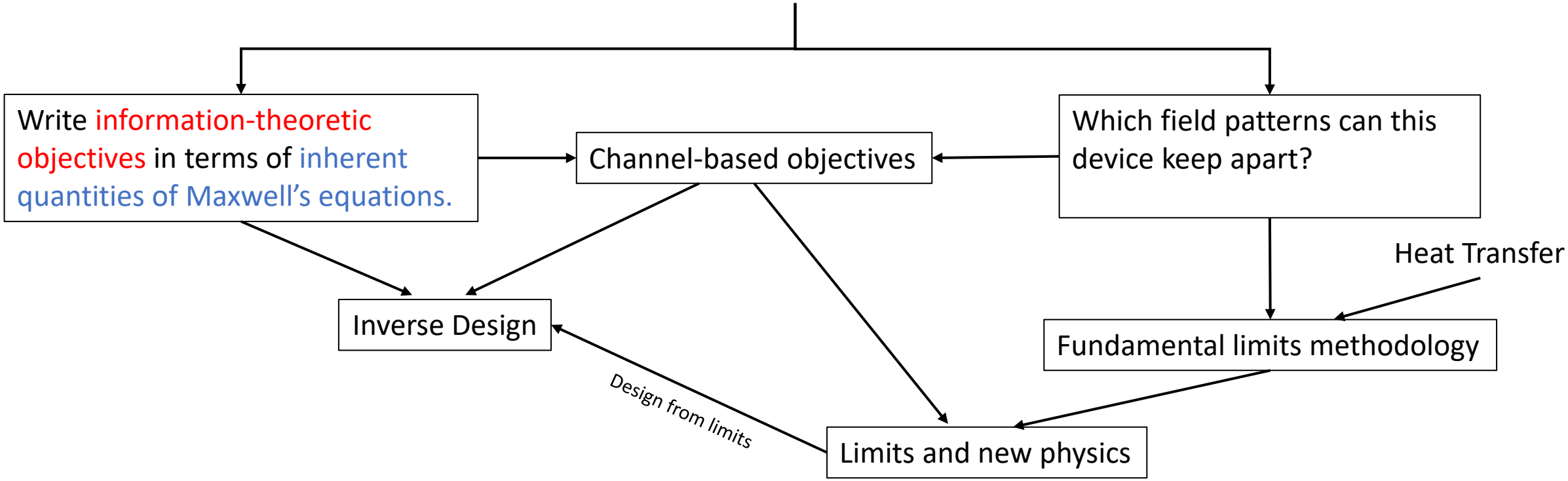
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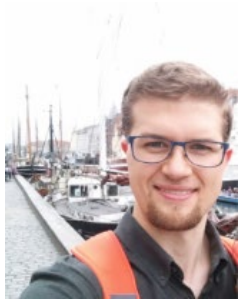
Summary of Part 2

We should be directly maximizing information-theoretic quantities for communication and sensing



Thank you for listening and thank you to

Prof. Sean Molesky



Paul Virally



Dr. Pengning Chao



Prof. Zin Lin



@ Polytech Montreal

@ MIT

@ Vtech

Collaborators

Prof. Alejandro Rodriguez



Thomas Maldonado



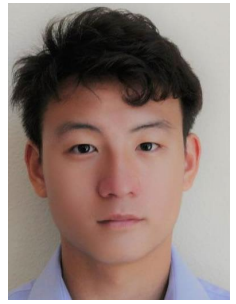
Dr. Benjamin Strekha



Dr. Jewel Mohajan



Francis Chen



RodPod @ Princeton

+ Links to code

I'm on the job market!

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alessioamaolo.github.io

CV

Physical Limits in Photonics: from Radiative Heat Transfer to Information Capacity and Design

Alessio Amaolo
Princeton University

Rodriguez Group @ Princeton

Prof. A. Rodriguez



T. Maldonado



F.C. Chen



Dr. B. Strekha



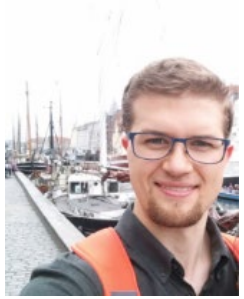
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Prof. S. Johnson

